AI-Powered Workflow Design for Purchase Ledger Data Preparation

Knowledge Transfer and Value Co-Creation through Human-AI Collaboration

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Abstract

Introduction The large-scale collection of public sector spending data by aggregators often yields records rife with severe semantic ambiguity, inconsistent formatting, and missing identifiers. Transforming this heterogeneous data into reliable business assets is a knowledge-intensive process that relies heavily on the tacit judgment of data specialists. Although artificial intelligence (AI) in data curation, human-in-the-loop (HITL) oversight, and knowledge transfer have been studied independently, there remains a gap in understanding how an integrated artefact can jointly support the preparation of highly ambiguous, knowledge-embedded financial data.

Research Question This thesis investigates how an AI-powered workflow can be designed to support knowledge transfer and value co-creation in data preparation. Specifically, it addresses two sub-research questions: (Sub-RQ1) how data specialists' tacit knowledge can be effectively externalized into cognitive AI nodes, and (Sub-RQ2) what specific dimensions of value are co-created through human-AI collaboration relative to manual preparation baselines.

Method Applying a Design Science Research (DSR) framework merged with Action Research (AR), this study was conducted in collaboration with Tendium AB, a Swedish data aggregator. Using the Framework Method, internal documents and raw datasets were analyzed to diagnose workflow bottlenecks and elicit system requirements. An AI-powered workflow artefact was subsequently designed and instantiated within the n8n orchestration platform. This artefact effectively combined deterministic processing, AI-enhanced contextual reasoning, and a defensive human-in-the-loop (HITL) governance gateway. The artefact was evaluated through a controlled laboratory experiment utilizing 20 stratified historical datasets from public sector entities in Sweden and Finland.

Results The implementation of the artefact significantly improved processing efficiency, reducing the average processing time per dataset from 2 minutes 41 seconds to 2 minutes 18 seconds. The more profound impact was that the human active attention time reduced to zero for 19 out of 20 datasets. Furthermore, the system reached 100% accuracy in routing structurally corrupt datasets to manual review and successfully resolved complex, multi-row accounting anomalies that typically resist rule-based automation. The study identified three core dimensions of co-created value: cognitive offloading, organizational knowledge retention, and risk mitigation using human-AI agency distribution.

Discussion The findings show that tacit data preparation expertise can be effectively codified into cognitive workflow nodes when safeguarded by a defined HITL boundary. By connecting previously isolated research streams, this study delivers a domain-specific architectural pattern and extends value co-creation theory into company business operations. Limitations of the study include a small evaluation sample size, a single-case context, and a restriction to structured file formats. Future research needs to explore broader generalization, the integration of additional file types, and the longitudinal effects on organizational learning.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
GPU	Graphics Processing Unit
LLM	Large Language Model
AP	Accounts Payable
SaaS	Software as a Service
RPA	Robotic Process Automation
IPA	Intelligent Process Automation
HITL	Human-in-the-loop
KIP	Knowledge-Intensive Process
KIT	Knowledge-Intensive Task
VCC	Value Co-creation
SDL	Service-Dominant Logic
IA	Intelligence Augmentation
ANT	Actor-Network Theory
KM	Knowledge Management
SECI	Socialization, Externalization, Combination, and Internalization
GenAI	Generative AI
HAIC	Human-AI Collaboration
TSEK	Task-specific expert knowledge

DSR	Design Science Research
AR	Action Research
BI	Business Intelligence
GT	Grounded Theory
RCA	Root Cause Analysis
LOA	Levels of Automation
SET	System Execution Time
AAT	Active Attention Time

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1

Introduction

1.1 Background

Nearly a decade ago, when scientists at Google [2] introduced the Transformer model, the world entered a period of rapid growth marked by intense competition among tech giants to develop increasingly powerful Large Language Models (LLMs). Since then, Artificial Intelligence or "AI" has become a ubiquitous buzzword for reasons both positive and negative. However, it is clear that businesses are now looking towards AI to solve existing problems and enhance current practices, such as increasing worker productivity.

One of the most significant leaps for businesses is AI's enhanced problem-solving capability. Companies have found ways to accelerate claims processing, prevent money laundering, and even replace legacy applications with systems that outperform human benchmarks [3]. The potential appears limitless, as different AI technologies are meant for specific tasks, such as utilization in data-related functions.

For modern businesses where data is a primary strategic resource, LLMs have exhibited outstanding results in data preprocessing [4]. Their adaptability permits them to parse text through multiple data formats efficiently, and in fact, dedicated models like TableLLM have been developed specifically for structured data manipulation [5]. This represents a systemic transition, as data preparation and cleaning traditionally consume over 80% of a data practitioner's time, according to the Anaconda 2024 State of Data Science report [6].

The reason why for instance, cleaning takes a long time is that data quality faces problems both on the instance level and schema level, as well as whether the data is from a single source or multiple sources. [7]. Hence, this becomes a manual and repetitive task.

When data is collected, cleaned, and transformed, it has gone through multiple steps to reach a specific outcome. But there are other parts of the process that a human is still doing. A company might then monetize this data or even develop data-based business models because when you take data from a raw format and aggregate it, its value inherently increases.[8, 9] This kind of business model is called a data aggregator which usually, according to Thomas & Leiponen [8], focuses on a specific sector. To support this, there are supporting activities and not just data cleaning. That is only one aspect for the data to become valuable.

1.2 Research Problem and Question

Despite the presence of AI, many of its possibilities remain untapped. A significant risk arises when AI excels at highly specific tasks but fails to grasp a wider context [3]. This limitation is reflected in the lack of research exploring AI development for capturing tacit knowledge in highly specific tasks. While AI can conduct complex operations, human monitoring remains central to business operations, stressing the need for tacit knowledge and contextual sensitivity [10]. This is especially true in regulated environments, where delegating decision-making to a mathematical model is rarely straightforward. Those shifts in responsibility raise critical questions regarding accountability when systems inevitably fail or produce unintended consequences [11].

While extensive research has addressed specific elements of data curation, such as automating basic tasks [12], comprehensive approaches to the entire curation pipeline remain in short supply[13]. This is largely due to the highly contextual nature of data needs, formats, and use cases. Because data is so heterogeneous, there is no “one-size-fits-all” solution. This variety complicates not only the end-to-end curation process but also the automation of its individual components, such as data cleaning [14].

Because current existing research treats AI automation, HITL oversight, and tacit knowledge as separate domains, it fails to provide comprehensive solutions for complicated data curation. Addressing this intersection

is critical to understanding how human-AI collaboration can safely manage ambiguous, knowledge-embedded datasets such as purchase ledger records.

To explore this gap, we present a main research question (RQ) with two sub-RQs.

Main RQ: How can an AI-powered workflow be designed to promote knowledge transfer and value co-creation in data preparation?

Sub-RQ1: How can data specialists' tacit knowledge be effectively externalized and integrated into the cognitive nodes of an AI-powered workflow?

Sub-RQ2: What are the specific dimensions of value co-created through this human-AI workflow compared to manual data preparation methods?

1.3 Collaboration with Tendium AB

Tendium AB is a Swedish Scale-up focusing on public tender management. The solution they offer is a Software-as-a-Service (SaaS) platform for discovering, analyzing, and winning government contracts. [15] Tendium also acts as a data aggregator, identifying tenders across Europe with a heavy presence in the Nordics and selling this data to the platform, which organizations that want to compete and sell to the government use. In this study, we're looking at one instance of Tendium's data preparation workflow, focusing on purchase ledger data.

1.4 Delimitations

The data curation workflow for payment data at Tendium is a complex, multi-stage process engaging various stakeholders, environments, and tools. This study does not cover the entire workflow; rather, it focuses specifically on the data preparation and curation stages prior to further ingestion. Consequently, the scope is limited to data handling, cleaning, and formatting, explicitly excluding data sourcing and downstream processing (e.g., ETL pipelines). Furthermore, the analysis is restricted to structured data files, meaning data retrieved from APIs falls outside the scope of this research. Specifically, the study concentrates on individual files in .csv and .xlsx formats.

1.5 Use of AI tools in this thesis

During the research, development of the artefact, and writing of this thesis, AI tools, specifically Google Gemini and Claude, were used to augment various aspects of the project. In accordance with academic transparency guidelines, the specific applications of these LLMs in this thesis are explained in this section.

AI models were used to proofread and revise manuscript drafts, enhancing grammatical correctness, readability, and academic cohesion. In addition, they provided technical assistance during troubleshooting and formatting adjustments in the LaTeX (Overleaf) environment.

The AI tools also served as technical co-pilots during the artefact's instantiation. This included providing architectural guidance on using n8n platform, drafting and debugging programmatic scripts for deterministic execution nodes, and systematically refining and fine-tuning the system prompts for AI cognitive nodes.

Moreover, during the evaluation phase, AI assisted in processing and calculating large volumes of historical performance data to generate structured quantitative tables.

While these LLMs significantly accelerated both the engineering and writing processes, it is vital to acknowledge their performance limitations. Because AI tools typically process individual prompts as isolated tasks without a complete understanding of the entire thesis context, their generated outputs can occasionally conflict with the overarching research narrative.

In line with the core HITL philosophy advocated in this study, AI was used strictly as an augmenting tool rather than as an autonomous creator. All generated code, prompt logic, statistical calculations, and revised paragraphs were subjected to thorough human verification and final governance by the authors to ensure absolute academic and technical accuracy.

2

Extended Background

The following sections outline the extended background of this research, detailing its key concepts, theoretical foundation, and related work. It begins with an introduction to the core concepts that are pivotal to the study. Then, it establishes the theoretical foundation that frames the research questions and guides the overall analysis. Lastly, the relevant prior literature for this study is discussed to contextualize the work.

2.1 Key Concepts

2.1.1 Data Curation and Purchase Ledger Data

Data Curation. In the field of business intelligence and data services, data aggregators collect and repurpose public or government datasets to provide value-added insights to the market [16]. However, as Loshin [16] notes, data obtained from external sources, whether public or independent, inevitably suffer from quality and consistency issues. These datasets must be processed and repaired within the aggregator's own organization. Transforming these raw, heterogeneous data into reliable business assets requires a rigorous "Data Curation" process. As defined by Stonebraker et al. [13], data curation is an end-to-end systematic engineering process that includes cleaning and transformation, semantic integration, and deduplication. Given the massive scale of external data, purely manual curation is no longer sustainable. Stonebraker et al. [13] explicitly argue that large-scale data curation "must adopt machine learning methods, supplemented

by human assistance when necessary," which provides the theoretical justification for this study's use of AI-powered workflows.

Despite the introduction of machine learning, data curation remains more than a simple IT automation task. Parmiggiani and Grisot [17] emphasize that data curation should be viewed as an "ongoing, context-dependent governance practice." When dealing with disparate open data, machines often struggle to handle every edge case independently. The curation process requires constant "human judgment" and negotiation to ensure deep data quality and logical soundness. This highlights that high-quality data curation is, by necessity, a dynamic process of deep human-machine collaboration.

Public Sector Purchase Ledger Data. In this study, the primary data sources for curation and preparation are purchase ledger records from public sector authorities in the Nordic region. A purchase ledger is, by definition, a sub-ledger used to track and manage purchases. It centralizes all purchases in one location as a record of the amounts an entity, e.g., a municipality, spends with its suppliers.[18] It works hand in hand with Accounts Payable (AP), showing which purchases have been paid for and which remain outstanding.[18] A typical transaction entered into the purchase ledger system first records an account payable, followed at a later date by a payment transaction that eliminates the payable balance.

In the public sector context, these ledgers allow public entities to track, account for, and report on specific financial flows, providing a distinct accounting identity for each spending agency.[19] Schaeffer defines the AP data housed in these ledgers as core business elements, including purchase orders, invoices, master vendor files, and payment records.[20] While an ideal purchase ledger requires highly structured data, data in the real world is often quite different.[21] A detailed purchase ledger contains massive arrays of data fields, such as supplier invoice numbers, identifying codes for items purchased, sales tax paid, and cost center details, which can vary widely in format and consistency. Schaeffer reveals the inherent gaps and ambiguities in practical AP and ledger systems, such as non-standardized invoice numbers and conflicting information in vendor master files. [21]

When data aggregators collect data from these purchase ledgers on a large scale from independent public systems across different municipalities, this inherent heterogeneity is exponentially amplified and leads to severe semantic ambiguity. For example, an isolated date field might lack context (making it impossible to distinguish between an invoice date and a

payment date), or similar payment amounts might represent either installment payments or duplicate records caused by the lack of uniform invoice numbers.

Within the overarching practice of data curation, this study focuses on the data preparation phase, where semantic ambiguities in purchase ledger data are resolved before integration, which constitutes the major bottleneck. Solving the logical and semantic ambiguities goes far beyond the scope of traditional rule-based formatting or alignment. It requires a system capable of contextual awareness and reasoning. This trust in "human assessment" and "business intuition" confirms that this data pre-processing stage is not a mechanical, manual labor task, but an exemplary knowledge-intensive task.

2.1.2 AI-Powered Automation

Traditionally, the automation of business processes has relied heavily on Robotic Process Automation (RPA) or hard-coded rules. These technologies operate on a deterministic rule-based logic [22]. Although highly effective for standardized tasks, in actual situations, they are inherently fragile when confronted with semantic ambiguities and missing values. To overcome these limitations, the application of AI in knowledge work is undergoing an essential shift from rule-based "reactive tools" to Intelligent Process Automation (IPA) and cognitive workflows [23].

Instead of depending exclusively on static "prompt-output" mechanisms or rigid IF-THEN rules, modern cognitive workflows integrate LLMs to handle complicated reasoning. Within this new mechanism, the system leverages what scholars describe as "agentic capabilities" [24]. Although not operating as a fully autonomous, fully unsupervised entity, AI nodes within the workflow exhibit adaptive and goal-oriented behaviors characteristic of agentic systems [25, 26]. They are capable of performing complex cognitive tasks, such as contextual disambiguation and causal reasoning, with little human intervention at the execution level.

By embedding these agentic capabilities within a structured workflow, the system achieves a collaborative Human-in-the-loop (HITL) architecture [27]. To clarify, the HITL concept discussed in this study is not the technical mechanism used in traditional machine learning solely for data annotation and model training. Instead, from the perspective of Information Systems Management, this study conceptualizes HITL as a human-AI collaborative workflow, a practical application of hybrid intelligence [28].

Human experts orchestrate workflow boundaries by delegating the cognitive burden of data cleaning and systematic reasoning to AI nodes. Consequently, the system generates highly standardized outputs, thereby liberating human data workers from micro-level error correction. This transformation establishes a solid foundation for high-level data authenticity verification and strategic human intervention in data preparation processes. In this approach, the AI is no more a mere execution tool but a digital collaborator with domain-specific reasoning capabilities, offering a compelling empirical context for exploring value co-creation in human-AI systems.

2.1.3 Knowledge Intensive Tasks and Tacit Knowledge in Data Preparation

Data Preparation as a Knowledge-Intensive Task. Since Blackler's [29] foundational study, knowledge work has been widely defined as an activity that heavily relies on experience, environmental comprehension, and intellectual effort. In the context of this study, the "data curation" lifecycle, which transforms massive, heterogeneous public AP data into high-value business assets, represents a typical Knowledge-Intensive Process (KIP) [30]. However, when focusing on the front end of this process (the "data preparation" stage), the specific actions of data cleaning and repair can no longer be broadly described as a macro-level process. Instead, they must be precisely defined as a series of micro-level Knowledge-Intensive Tasks (KITs).

Heravizadeh et al. [31] define KITs as complex tasks characterized by high uncertainty; they cannot be fully resolved using predefined algorithms and require operators to actively apply professional training, experience and judgment. When processing AP data, actions such as determining the correct assignment for missing codes or resolving the ambiguity of isolated dates are prime examples of KITs. As van Elst et al. [32] point out that traditional, highly structured workflows often fail when confronted with KITs because they are only designed to handle predictable, deterministic decisions and cannot respond flexibly to specific business contexts.

The Role of Tacit Knowledge in Data Preparation. When executing these knowledge-intensive data preparation tasks, the core resource driving correct decisions is not a written manual, but the tacit knowledge residing in the workers' minds. According to Smith [33], unlike "explicit knowledge" (such as static vendor code tables) that can be easily codified and stored, tacit knowledge is rooted in human action, accumulated experience, and specific business contexts. For instance, when processing public

procurement data across different municipalities in Finland, data specialists or account specialists can accurately discern whether an isolated date represents an invoice date or an actual payment date. Furthermore, based on their understanding of local payment habits, they can infer whether multiple similar payment amounts indicate a duplicate payment or a refund. These kind of heuristic judgments and intuitive understandings of data context are concrete manifestations of tacit knowledge in real-world business environments.

The highly personalized and context-dependent nature of tacit knowledge creates the most significant bottleneck for traditional automation technologies during data preparation. Because tacit knowledge is extremely difficult to codify directly into traditional "IF-THEN" software instructions [33], these knowledge-intensive tasks remain trapped in manual processing. This is not only time-consuming but also leaves organizations vulnerable to the loss of intellectual assets due to employee turnover.

Therefore, the core challenge in achieving intelligent process automation is overcoming the limitations of traditional RPA to "externalize" and "transfer" human experts' tacit reasoning to AI nodes equipped with contextual understanding. This technical requirement establishes the essential foundation for subsequent discussions on achieving Knowledge Transfer and Value Co-creation between humans and machines through AI-powered workflows.

2.2 Theoretical Foundation

2.2.1 Value Co-creation with a Service-Dominant Logic Perspective

Value Co-creation (VCC) represents a fundamental shift from traditional business models. Traditional Goods-Dominant Logic assumes that value is unilaterally created by the companies and delivered directly to passive recipients. In contrast, the VCC perspective posits that value is jointly created by multiple actors within an ecosystem through resource exchange and interaction. The theoretical cornerstone of this concept is the "Service-Dominant Logic" (SDL) proposed by Vargo and Lusch [34]. Within the SDL framework, Vargo and Lusch [34] make a distinction that is crucial for this study: the difference between operand resources (e.g., physical products or raw data) and operant resources (i.e., knowledge, skills, and core competencies). SDL perspective emphasizes that the essence of value creation

is not the processing of materials, but rather the process by which different parties apply their operant resources to provide services to one another. Subsequently, Ranjan and Read [35] operationalized the abstract concept of VCC into two specific dimensions: co-production and value-in-use. Furthermore, Grönroos [36] emphasizes that genuine co-creation must be built upon substantive "interaction" mechanisms between the involved parties.

In traditional SDL research, the primary actors in VCC are typically assumed to be limited to "human-to-human" interactions between companies and customers. However, with the rapid advancement of AI technologies, the VCC theory is undergoing a profound transition. Barile et al. [37] note that in the digital age, through "Intelligence Augmentation" (IA), AI technologies have evolved from passive tools into active co-creators capable of participating in resource integration. Furthermore, in the systematic literature review of Kaartemo and Helkkula [38] advocate for the application of Actor-Network Theory (ANT), recognizing that technological entities, such as AI and algorithms, also possess "agency" in VCC. This evolution signifies that the research context of VCC has officially expanded from interpersonal interactions to the boundary of "human-to-non-human" interactions [39]. In this new context, machines are no longer mere containers for processing operand resources, they have become digital collaborative partners capable of contributing operant resources.

Mapping these theories onto the empirical context of this study, specifically, the intelligent preparation of AP data, reveals that processing ambiguous data is inherently a deep "co-production" process. Within these micro-level knowledge-intensive tasks, neither humans alone nor LLMs alone can independently generate high-quality data outputs. Therefore, the AI-powered workflow developed in this study demonstrates a classic human-AI VCC mechanism. Human data specialists act as the leading actors, contributing their deep tacit business knowledge and workflow boundary design (human operant resources).

Simultaneously, the AI nodes act as collaborative actors, contributing their robust contextual parsing and causal reasoning capabilities (AI operant resources). As Zhang et al. [40] note in their discussion of agency distribution in human-AI co-creation that clearly divided collaboration maximizes the respective strengths of both parties.

However, for this human-AI VCC mechanism to function smoothly, especially regarding the transfer of human operant resources to the AI agent,

tacit knowledge must be effectively externalized and transmitted. This inevitably introduces another core theoretical foundation of this study: Knowledge Transfer.

2.2.2 Knowledge Transfer and the SECI Model

When discussing AI-related system design, the term "Knowledge Transfer" is easily confused with the concept "Transfer Learning", referring to the transfer of underlying algorithmic weights in the field of machine learning. However, in this study, the theoretical foundation of knowledge transfer is firmly rooted in the discipline of Knowledge Management (KM). According to Argote's [41] classic definition, knowledge transfer is the process through which experience or knowledge moves from one entity to another, substantially influencing the recipient's behavior. Within the context of data preparation in this paper, this transfer refers to specifically describe the process of extracting the "tacit knowledge" used by data specialists to resolve data ambiguities and effectively transmit it to the digital workflow system. This transfer provides the system with the necessary cognitive processing capabilities.

The core mechanism of organizational knowledge transfer can be thoroughly analyzed through the classic SECI (Socialization, Externalization, Combination, and Internalization) model proposed by Nonaka [42]. Within this framework, the process of Externalization is considered the most critical yet challenging stage, in which tacit knowledge is converted into explicit knowledge. In the era of traditional information technology, because tacit knowledge relies heavily on intuition and context, externalizing it into deterministic computer code was nearly impossible. This bottleneck meant that traditional automation systems could only passively store explicit data. They were unable to inherit the deep, tacit judgments of experts, thereby creating an insurmountable human-machine "knowledge gap" when handling complex tasks.

However, with the introduction of Generative AI (GenAI), such as LLMs, the traditional SECI model is experiencing an essential shift. As Böhm and Durst [43] highlight, in the GenAI era, machines are no longer merely "passive databases" for storing rigid rules. Instead, through their strong natural language understanding and contextual reasoning capabilities, they have evolved into active recipients and generative entities within the knowledge transfer process. Falckenthal et al. [44] further confirm that utilizing AI to capture and transfer human tacit experience has become an effective approach to mitigate organizational knowledge loss.

Although AI lowers the barrier to externalization of tacit knowledge, in highly compliant data curation tasks, this knowledge transfer is by no means a fully automated black-box operation. Rather, it heavily relies on HITL as a Hybrid Intelligence architecture, which denotes the "cognitive guidance, interpretation, and constraint" [45] that human experts apply to AI outputs within a specific business context. In the AI-powered workflow developed for this study, human experts effectively "externalize" their tacit judgment logic to the AI by orchestrating workflow nodes and crafting System Prompts embedded with business rules. The AI nodes then interpret these instructions to execute complex data cleaning, which represents "internalization" at the machine level [46] [40]

This mechanism not only closes the knowledge transfer loop within the human-AI system, but also provides the indispensable driving force for achieving the Human-AI Value Co-creation discussed previously.

2.3 Related Work

2.3.1 AI-Empowered Automation of Data Curation Workflows

In recent years, researchers have widely explored the potential of AI in enhancing large-scale data quality and optimizing data governance. Sugureddy [47] notes that integrating generative AI into data governance frameworks helps data stewards effectively streamline data classification, metadata management, and policy execution, thereby improving overall data quality. Within specific vertical domains, de Zegher et al. [48] examined the application of AI-based data curation in healthcare, demonstrating AI's critical role in processing heterogeneous data to build a patient-centric Health Data Space. However, while these studies establish AI's value in macro-level data governance and large-scale dataset integration, they primarily focus on system architecture. They pay relatively less attention to the micro-level execution mechanisms of underlying data preprocessing and the cleaning of raw data.

In the stage of data cleaning and preparation, the technological approach is undergoing a significant shift. Chai et al. [49] emphasize that machine learning and AI technologies are increasingly applied to automate data preparation tasks. Addressing this shift, Leoi [46] conducted a key comparative study, finding that while traditional rule-based data cleaning systems provide reliability and transparency, they often lack adaptability when processing large, complex, unstructured, or constantly evolving

datasets. In contrast, AI-driven technologies excel in scalability and flexibility. Cherekar [50] further supports this view, noting that using AI technologies for automated error detection and data cleaning not only significantly improves data reliability but also substantially reduces tedious manual labor. This has been validated in practice within fields such as finance and Business Intelligence.

Although AI demonstrates advantages in isolated cleaning tasks, data preparation in the real world for unknown datasets is rarely a one-time process. Mahdavi et al. [14] explicitly note that it is impractical to define all necessary data quality constraints at once for unfamiliar datasets. Instead, users must implement iterative cleaning workflows to determine the correct combination of cleaning procedures. Furthermore, in highly specialized and complex domains, such as finance and public administration, pure automation often faces significant limitations. When discussing the integration of AI and RPA in financial accounting, Zhan et al. [51] point out that while combining these two improves efficiency, traditional automation still exhibits clear shortcomings in handling unstructured data and complex decision-making scenarios. To bridge this gap, Peña et al. [52] strongly advocate for a HITL mechanism in their study on data curation for public affairs documents. They emphasize that when dealing with complex documents characterized by high variability and multi-modal features, systems must incorporate human-centric machine learning approaches to comprehend complex information layouts that purely text-based methods often miss.

2.3.2 Knowledge Transfer in AI-Empowered Workflows

In recent years, academia has widely recognized the potential of AI in overcoming traditional KM bottlenecks. Jarrahi et al. [53] note that AI and humans are forging a novel partnership in the creation, storage, and application of knowledge. Furthermore, a large-scale empirical study by Standaert and Andries [54] confirms that deploying AI technologies can effectively mitigate human limitations regarding ability and motivation during the knowledge transfer process. Especially in organizational contexts with limited opportunities for interpersonal interaction, AI acts as a critical lever to drive the flow of tacit knowledge and process innovation.

However, considering AI as a black-box tool to replace human task execution exposes organizations to a severe risk of knowledge loss. Scibelli and Stevens [55] issue a strong warning in their recent study: when organizations blindly adopt applied AI systems in knowledge-intensive roles,

the traditional exchange of tacit and implicit knowledge from experts to successors is severed. The oversimplified conversion of human practices into machine functions blurs the human-machine boundary, ultimately "breaking the chain of knowledge transfer." [55] This implies that fully automated systems, completely removed from human involvement, not only fail to inherit the deep business logic of experts but also hinder the future evolution of domain expertise.

To address the aforementioned knowledge loss crisis, academia has increasingly turned to Human-AI Collaboration (HAIC) mechanisms to facilitate secure and sustainable knowledge transfer. Spitzer et al. [56] highlight the importance of preserving expert experience at the micro-task level. They proposed an HAIC framework that aims to effectively externalize and transfer task-specific expert knowledge (TSEK) to novices via AI systems. Within the broader scope of knowledge organizations, Cheng et al. [45] explicitly advocate for the integration of HITL mechanisms. They emphasize that, given the challenges concerning the accuracy and transparency of AI outputs, the core role of human experts has shifted toward guiding, interpreting, and validating AI systems. Similarly, Falckenthal et al. [44] confirm that the use of AI in conjunction with human expert guidance represents a cutting-edge approach to capture and transfer intergenerational tacit experience.

2.3.3 Human-AI Value Co-creation in Practice

Research on the role of AI in VCC has increased significantly. AI now act as active agents, substantially participating in resource integration and value creation [38]. However, the majority of existing applied research tends to anchor its perspective on macro-level service ecosystems or customer-facing, front-end interactions. For instance, Leone et al. [57] explored macro-level digital transformation and AI value co-creation mechanisms within B2B healthcare ecosystems. Similarly, Kot et al. [58] focused on customer-facing scenarios, analyzing the value co-creation dimensions and service attributes of conversational agents replacing traditional customer service representatives.

While these studies validate AI's potential to drive value co-creation, their emphasis on external ecosystems and front-end service interactions has led the academic community to relatively overlook the organizational back-office. Specifically, there is a lack of focus on how human-AI co-creation

mechanisms are practically implemented during highly complex and tedious internal knowledge-intensive tasks, such as the cleaning and preparation of massive datasets.

To address the limitations of these aforementioned application scenarios, some scholars advocate shifting the research focus toward the internal micro-foundations of collaboration within organizations. Raftopoulos and Hamari [59] emphasize that to translate AI's technological potential into sustainable business value, organizations must build systematic internal microfoundations. This requires a strategic shift from blind machine replacement to human augmentation. Similarly, Barile et al. [37] argue for adopting an "Intelligence Augmentation" (IA) perspective in the digital age. They suggest viewing AI as a co-creation tool that amplifies and extends human expertise, rather than merely an automated execution engine. However, while these studies establish the core role of augmented intelligence in internal VCC at a strategic level, they largely remain conceptual. They have yet to detail the fine-grained collaboration and division of labor required between humans and machines when executing specific, complex data tasks.

To operationalize this augmented co-creation mechanism at the micro-task level, recent research has shifted its focus toward task characteristics and the distribution of human-AI control. Braathen and Findsrud [60] proposed a taxonomy for human-AI VCC, explicitly stating that whether a system adopts an "Automation" or "Augmentation" model fundamentally depends on the core dimension of the complexity of task-in-context. This implies that for highly complex tasks with strong temporal and spatial dependencies, such as the AP data addressed in this study, which lacks standardized structures and is fraught with semantic ambiguity, simple automation is bound to fail. Instead, a deep, augmented co-creation model must be applied.

Furthermore, Zhang et al. [40] review top HCI literature and note that the core of high-quality collaborative co-creation lies in the appropriate distribution of agency. Previous best practices at the micro-level indicate that in complex digital collaborations, human experts should retain agency over macro-level boundary setting and system guidance, while delegating agency for micro-level computational analysis and pattern recognition to the AI. This recent theoretical insight regarding agency distribution provides direct practical endorsement and theoretical justification for the system architecture adopted in this study, wherein human experts orchestrate workflow rules and AI nodes execute underlying data reasoning.

2.3.4 Research Gap

In summary, while existing literature provides a robust foundation for this research, it also reveals a significant intersectional gap. Previous studies have demonstrated the superiority of AI in automated data preparation workflows (Section 2.3.1), highlighted the necessity of HITL mechanisms to prevent the loss of tacit knowledge (Section 2.3.2), and established the principles of agency distribution for micro-level human-AI VCC (Section 2.3.3).

However, these three domains have largely evolved in isolation. To date, there is a distinct lack of empirical research that integrates these dimensions into a unified, practical artefact, specifically within the context of data curation workflows. Existing studies have yet to demonstrate how a cognitively empowered, HITL workflow can be designed to concurrently externalize tacit knowledge and co-create value when processing highly ambiguous, knowledge-embedded financial datasets, such as AP data in this study.

This study aims to address this precise gap. By designing and evaluating an AI-empowered cognitive workflow artefact for AP data cleaning, this research translates the theoretical microfoundations of human-AI collaboration into a tangible, domain-specific solution, the design logic of which will be detailed in the following chapters.

3

Methodology

The following section describes the study's methodological approach. It begins with a discussion of the qualitative research design and the rationale behind the chosen method. Next, the data collection methods are presented. Finally, ethical considerations relevant to the study are discussed as well as the study's credibility.

3.1 Design Science Research

In order to explore the dynamics of human-AI value co-creation within data curation, this paper adopts a Design Science Research (DSR) approach. DSR inherently is about problem-solving [61] by designing and developing an artefact, a solution for a problem and evaluating it.

In chapter 2, we discussed problems with data obtained from external sources and how many lack quality and consistency. When it comes to purchase ledger data across different municipalities, we are dealing with datasets from different Information Systems. As stated earlier, this leads to severe semantic ambiguity in files because there is no a one format to be agreed upon. Because of this kind of heterogeneity in data, this presents a structural choke point, so merely observing or describing the phenomenon of semantic ambiguity is insufficient.

This study uses a DSR process proposed by Johannesson & Perjons [1]. It consists of 5 activities to guide the researcher throughout the design science project. The following picture is a visualization of the whole process.

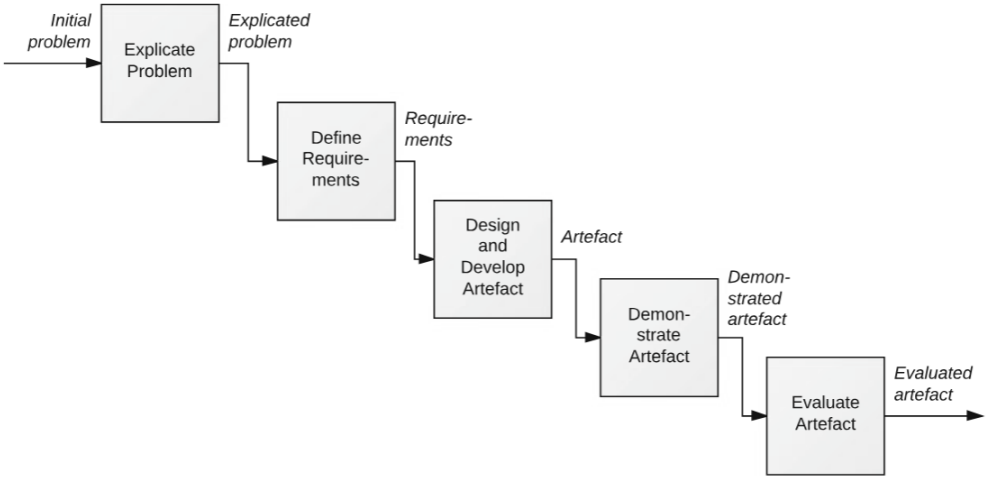


Figure 3.1: Overview of the DSR framework [1]

Explicate Problem: This stage sets the stage for the rest of the design science project. In this paper, this means analyzing existing problems within data curation workflows, focusing mainly on data cleaning and data preprocessing. As the focus is on purchase ledger data, it is important to find specific problems related to it. To help with problem explicating, sub-activities are used to be as clear as possible. These sub-activities are defining the problem precisely, positioning and justification of the problem and finding its root causes. [1]

Define Requirements: As a problem has been explicated in the previous stage, it is time to elicit requirements for the proposed artefact. Essentially, this is just a continuation of the problem explication, this time using it as a guideline for design and development. A requirement here is a property of the artefact [1], which the stakeholders can assess as a useful attribute that needs to be solved. Categorizing the collected requirements will help as well with prioritizing which this artefact can contribute to the most. If a requirement is expressed but doesn't demonstrate value, it shouldn't be given high priority.

Design and Develop Artefact: This is the part of the design science project where a researcher gets to create the artefact. The artefact should address the explicated problem and fulfill the defined and chosen requirements.[1] In this study this means that the artefact is built using appropriate tools.

Demonstrate Artefact: A designed and developed artefact has no meaning if it isn't demonstrated. This means that the questions of how the artefact works and why the artefact works should be answered. [1]

Evaluate Artefact: The last stage of this process is the evaluation, where the artefacts perceived usefulness will be measured.

DSR is the ideal methodological fit for this research because the objective is to solve a practical, real-world problem. The methodology guides the paper through different steps to iteratively build a tangible solution and to relate it to an existing knowledge base.[1]. The proposed artefact is a process automation prototype with the objective of improving and automating tasks within the data curation process by integrating AI into a currently manual process, only including a human. The proposed artefact is a process automation prototype with the objective of optimizing data curation by integrating AI into a manual and human-centric workflow.

3.2 Research Strategy

3.2.1 Action Research

DSR is compatible with various research strategies, and none should be prematurely excluded at the beginning of a project [1]. For this study, the chosen research strategy is Action Research (AR). AR emphasizes practicality, requires the active participation of practitioners, and aims to produce a tangible artefact of value for the specific environment in which the project is conducted [1]. This aligns closely with the present study, as one of the authors is a practitioner who is familiar with the workflow, and a primary goal of the thesis is to develop a working prototype. Because this study aims to directly improve organizational efficiency and effectiveness, it specifically works well with the goals of Technical Action Research [1].

This strategy is furthermore justified by the artefact's role as a dedicated data workflow within a highly specific organizational context. The data in question, AP records, primarily consist of structured data, in contrast to the unstructured nature of sources like social media. Because the data sources, transformation logic, and end-use integrations are highly tailored to this environment, the workflow is inherently distinct. Developing an effective solution therefore requires the hands-on, contextualized approach that AR provides.

A common challenge of AR is that findings are often tied to a single local practice, which can complicate generalizability [1]. This is true in

Tendium's case, as this specific workflow may not represent the most common method for preparing data for Business Intelligence (BI). However, the higher-level principles of data cleaning and preparation share fundamental characteristics across many organizations. The core research question of this study investigates how an AI-powered workflow can be designed to facilitate knowledge transfer and value co-creation in data preparation. AR provides the necessary practical framework to answer this question in situation. Any concerns regarding researcher impartiality or the limits of external validity will be examined in the discussion of the proposed artefact's results.

3.2.2 Alternative Research Strategies

Another consideration for a research strategy was the use of a case study. This approach enables a researcher to examine a single instance of a phenomenon, dive deeply into its real-world setting, and thoroughly consider the complex relationships and processes within it [1]. Applying this strategy would mean investigating the specific data workflow strictly as one instance of data preprocessing. While a case study requires a holistic view that captures all relationships within the workflow and its surrounding environment [1], it is traditionally an observational approach. Although investigating this unique data aggregator could offer broader insights into the value of human-AI workflows, a pure case study lacks the active intervention required to actually build and implement a working prototype. Furthermore, case studies ultimately suffer from the same criticisms as Action Research, as their results are often difficult to generalize to other instances.

To address the criticism of generalizability, a survey strategy was also evaluated. The notion behind a survey is to investigate the subject from a bird's-eye view, utilizing a much larger scope of data collection [1]. However, significant problems would arise in sampling. The data in question is distinct from other forms of data; its use specifications and the ultimate goals of the data cleaning process are highly specialized. Because this project focuses on one specific workflow, a survey would require access to numerous other data cleaning workflows within Tendium, which are not currently available. While a survey would yield results more suitable for generalization, it would come at the cost of finding a practical solution to a specific, applied problem. Because AR directly contributes to practical problem-solving and value creation, it remains the most appropriate research strategy for this study.

3.3 Data Collection & Analysis

This study will employ a multi-method qualitative approach for data collection. To map the data workflow and establish baseline objectives for the artefact, the process begins with documentary analysis. This involves reviewing publicly available information on the structure of the collected data, alongside internal organizational records, to understand how data is moved, by whom, and through what processes and how it is transformed to fit the schema. To address any gaps in the documentary evidence, semi-structured interviews will be conducted as a complementary method if necessary. This approach is particularly effective for capturing tacit knowledge as practical insights that would otherwise remain undocumented within the organization's Knowledge Management Systems (KMS).

A common challenge however with using document analysis as a primary data collection method is that establishing the credibility of the sources is not always straightforward [1]. However, within the context of AR, internal organizational records are inherently valuable because they accurately reflect the practical reality and current state of the organization being studied. Moreover, the publicly available information regarding purchase ledger data structures, such as open data published by public authorities, is generally recognized as highly trustworthy [1].

To analyze the collected data, this study employs the Framework Method as an analysis approach. The framework Method has become a popular and highly systematic approach in other fields though it originated in health research. One of the features is a matrix-based output consisting of rows, columns, and cells of summarized data, which provides a structured format to reduce and analyze extensive information.

The Framework Method was chosen for this study due to several benefits that align with the objectives of DSR. First, it is not strictly aligned with a particular epistemological, philosophical, or theoretical approach, making it an incredibly flexible tool. [62] For a DSR project aiming to solve practical organizational bottlenecks, this approach is beneficial as it allows the analysis to be guided by pre-existing theoretical constructs regarding knowledge-intensive tasks, while still leaving space to discover contextual nuances within the collected data.

Grounded Theory (GT) was considered as an alternative analytical strategy. It relies on open, inductive coding and the constant comparative method to develop theory directly derived from the data through a set of interconnected stages. While its purely inductive nature is excellent for allowing

research to remain highly receptive to emerging themes without the constraints of predefined categories, it was ultimately deemed unsuitable for the specific goals of this study. The primary reason for moving away is that its core focus is on generating new social theory. [1] The objective of this thesis is not to generate broad social theory but rather to systematically diagnose practical workflow inefficiencies, extract tacit knowledge, and elicit requirements for an AI-powered artefact. While the Framework Method can also greatly facilitate the "constant comparative" techniques popularized by Grounded Theory, it does so without the mandate of generating social theory. [62] Instead, it offers a more pragmatic, matrix-driven structure that is better suited for cross-referencing organizational documents, managing heterogeneous data sets, and supporting the highly structured outputs required for the artefact development in DSR. [62]

3.4 DSR Evaluation Strategy

This study will implement an ex-ante evaluation strategy. The evaluation of the artefact in this study is defined as an ex ante, formative evaluation. Ex-ante evaluations are ideal for quickly assessing an initial prototype, as they are inexpensive and, moreover, pose few to no ethical risks. [1] Moreover, the prototype will be run locally and not deployed fully which further speaks to ex-ante evaluation as opposed to ex-post where an artefact would be deemed to be fully deployed.

Because of the artefact being a locally developed prototype, and not fully deployed to data practicioners, a field experiment approach would be too resource-intensive and premature. Instead, this study will use a lab experiment approach. This will be a form of an artificial evaluation as the evaluation will be in a controlled setting and not in a real-world scenario, i.e. in actual day-to-day use. Artificial evaluations lack in suitability for studying the effectiveness of an artefact, meaning that it would be able to produce desired outcomes in a real-world setting. However, its low cost and ability to produce structural qualities make it a good fit for this study.[1]

By running a controlled set of 20 different data files through the prototype, the evaluation simulates real-world performance to investigate if the artefact can answer to the explicated problem, meet defined requirements and compare its efficiency against the baseline. These are the main goals of the evaluation; other goals include side effects and possibly identifying

future improvements for the artefact, as mentioned by Johanneson & Perjons. [1]

3.5 Ethical Aspects

Regardless of the chosen strategy, every rigorous research paper must carefully consider its ethical implications. In Action Research (AR), standard research ethics apply, requiring that permissions are obtained, confidentiality is maintained, and identities are protected [63].

With this in mind, this study ensures the strict confidentiality of any sensitive information regarding Tendium's business strategy. During the data collection phase, explicit permission to use any data will be secured. Furthermore, the development of the artefact will remain transparent and open to suggestions from the organization. Because the aim is to deliver a functioning prototype, continuous input from the company is both practically important and ethically sound.

To mitigate common criticisms directed at AR, this study will not attempt to generalize its results beyond this specific case; all discussions regarding the findings will be kept firmly within these boundaries. Secondly, AR presents unique ethical considerations regarding the balance between practical work and scientific research. To address potential issues related to time constraints, workload balance, and possible reduced compensation, this study will be localized and kept small-scale. Rather than attempting a broad investigation that could yield superficial outcomes due to time limitations, research integrity is maintained by focusing on a narrow, manageable scope.

4

Problem and Requirements

4.1 Process of Problem Analysis and Requirements Elicitation

As outlined in the methodology chapter, the initial data collection strategy proposed a multi-method approach, with semi-structured interviews planned as a complementary method to address any potential gaps in the documentary evidence. However, during the initial data collection phase, it became evident that interviews would not be necessary. Because one of the researchers was actively involved in data operations as a practitioner, the study gained exceptional access to a rich, extensive corpus of authentic operational documents and system logs.

A review of these internal archives, which included 11 core business documents and 10 empirical datasets extracted from the organization's internal knowledge management system (Confluence), shared collaboration drives (Google Drive), and cloud storage environment (AWS S3)—provided a complete and comprehensive understanding of the workflow. The collected materials successfully captured the necessary knowledge, workflow behaviors, and system bottlenecks, yielding all the required evidence. Consequently, the planned interviews were bypassed, and the problem diagnosis and requirements elicitation were conducted relying exclusively on the documentary analysis.

4.1.1 Multi-source Data Collection and Categorization

During the first two phases of DSR, this study extracted a rich corpus of empirical data from the organization's internal knowledge management system (Confluence), shared collaboration drives (Google Drive), and cloud storage environment (AWS S3). To adhere to academic standards, protect commercial confidentiality, and facilitate the analysis of internal documents using the Framework Method, the collected 11 core business documents and 10 empirical datasets were reclassified into four logically progressive categories. These documents were subsequently renamed and categorized for academic rigor as follows:

Category 1: Business Context and Process Baselines This category provides the macro-level background, policy environment, and existing workflow configurations of the data collection project. It serves as the foundation for understanding the entire "As-Is" process.

- *Workflow diagram* $\rightarrow$ **Current Data Processing Workflow**: Visually illustrates the current data routing paths utilized by the data team.
- *Project Description* $\rightarrow$ **Data Collection Project Description**: Defines the core objectives and business scope of the project.
- *Automation - Business Case* $\rightarrow$ **Automation Business Case**: Documents early internal organizational discussions and expectations regarding process automation.
- *National Breakdown (Finland) & (Sweden)* $\rightarrow$ **Public Sector Context Overview**: Provides the governmental administrative structure and policy background necessary to understand the sources of purchase ledger data.

Category 2: Data Characteristics and Anomaly Rules This category summarizes the tacit knowledge of the data practitioners. It meticulously records the complex characteristics of purchase ledger data and the empirical rules developed by the team to address these pain points.

- *Data Issues Documentation* $\rightarrow$ **Historical Data Issues Log**: Compiles the high-frequency formatting and logical errors encountered during the data cleaning process.
- *Accounting anomaly* $\rightarrow$ **Accounting Anomaly Resolution Guide**: Explains the underlying business logic behind financial phenomena,

such as invoice splitting and duplicate amount in a purchase ledger dataset. The goal with this guide is to reshape every accounting structures into one accepted structure, one row per invoice line per supplier.

- *Dictionary superfluous variables* → **Superfluous Variables Dictionary**: Defines the non-standard mapping relationships between original data fields and target requirements.
- *How to format files* → **Data Pre-processing SOP**: Describes the specific manual steps analysts take to format files in practice.

Category 3: System Constraints and Errors Feedback This category defines the output standards that the constructed artefact must meet and exposes the limitations of traditional automation tools when handling complex data.

- *Glue Flow Requirements* → **Downstream ETL Integration Requirements**: Strictly defines the formatting thresholds for inputting cleaned files into Amazon AWS Glue.
- *AWS Glue Flow Feedback and Errors* → **ETL System Error Logs**: Records data rejection and system error messages caused by incomplete preprocessing or structural anomalies.

Category 4: Raw Empirical Datasets Serving as the ultimate carrier for all aforementioned rules and processes, authentic business data samples were extracted directly from the company's AWS S3 data lake for low-level cross-validation and testing.

- *10 datasets of municipal purchase ledger data* → **Raw purchase ledger datasets from Finnish and Swedish Municipalities**: Contains snapshots of real accounts payable and payment records from various levels of local government in Finland (5 datasets) and Sweden (5 datasets).

By categorizing collected documents into four categories, this study not only ensures data triangulation for problem diagnosis but also establishes a solid empirical foundation, which facilitates the subsequent use of the Framework Method to precisely define the requirement boundaries for the artefact.

4.1.2 Application of the Framework Method

To systematically transform the heterogeneous internal documents and raw datasets into structured system requirements, this study adopted the Framework Method [62]. This method is particularly suitable for multi-disciplinary research as it allows for both deductive (theory-driven) and inductive (data-driven) thematic analysis. The process followed five distinct stages as detailed below.

Stage 1: Familiarization

The researcher achieved immersion in the data by repeatedly reviewing the 11 internal documents and 10 raw purchase ledger datasets. By cross-referencing the Data Pre-processing SOP (GDrive_2) with the Historical Data Issues Log (Confluence_2), it became evident that a significant "logic gap" exists between the messy reality of municipality data and the rigid formatting requirements of the downstream ETL pipeline.

Stage 2: Identifying a Thematic Framework

Drawing upon the theoretical foundations established in Chapter 2 (e.g., Knowledge-Intensive Processes and Tacit Knowledge), an initial thematic framework was developed. Three core themes were identified to categorize the practical problems:

- **Theme 1:** Friction between Data Quality and Rigid System Constraints.
  - *Focus: Discrepancies between raw input and AWS Glue requirements.*
- **Theme 2:** Reliance on Tacit Knowledge in Bridging the Quality Gap.
  - *Focus: Need for human judgment to resolve accounting errors and semantic ambiguities.*
- **Theme 3:** High Cognitive Load and Scalability Bottlenecks.
  - *Focus: Time-consuming manual formatting and the inefficiency of current human-led workflows.*

Stage 3: Indexing and Coding

Using the initial framework, the researcher performed line-by-line coding of the documents. Specific excerpts were labeled with micro-codes. For

example, the instruction in GDrive_3 stating that users must "use your own judgment to tell the computer which accounts you want to eliminate" was indexed under the code Reliance_on_Judgment within Theme 2. Statistics regarding time-loss in GDrive_1) were coded as Resource_Drain under Theme 3.

Stage 4: Charting and Developing the Analytical Matrix

The indexed data were summarized into an analytical matrix. Table 4.1 provides a summary of the mapping between key data sources and the three themes. As shown, some documents (e.g., Accounting Anomaly Guide) focus exclusively on cognitive judgment, while others (e.g., Business Case) highlight broader organizational inefficiencies.

Table 4.1: Analytical Matrix of Practical Problems (Summary)

Data Source	Theme 1: System Friction	Theme 2: Tacit Knowledge	Theme 3: Cognitive Load
Data Issues (Confluence_2)	Evidence of missing org-numbers and blank supplier names causing import failure.	Analysis of "cost-effective judgment" needed to exclude low-quality data.	–
ETL Feedback (Confluence_7)	Critical log entries showing "Job failures" due to minor formatting errors.	–	Time spent debugging failed AWS Glue jobs.
Accounting Anomaly (GDrive_3)	/	**Core Evidence:** Requirement for understanding Purchase ledger data structures	High effort required for cleaning these files.
Automation Case (GDrive_1)	/	50% of files require "direct follow-up" and legal/accounting expertise.	**Core Evidence:** 60-120 hours of labor per data collection project.
Formatting SOP (GDrive_2)	Requirements for dot decimal separators and semicolon delimiters.	/	Manual "Text-to-columns" and R-coding tasks for data prep.

Stage 5: Mapping and Interpretation

The final stage included a holistic synthesis of the analytical matrix to identify patterns and relationships between the identified themes. Rather than viewing data quality issues and system constraints separately, this stage focused on interpreting the "Cognitive Disconnect" revealed by the framework: the persistent reliance on human tacit knowledge to bridge the purchase ledger data between heterogeneous datasets and the deterministic requirements of the downstream ETL pipeline.

The interpretation of the matrix highlighted three critical insights:

1. **The Semantic Gap:** The most significant friction occurs when raw data (e.g., split invoices or missing identifiers) requires a level of domain-specific "sense-making" that the a rule-based automation cannot provide effectively.

2. **The Efficiency Ceiling:** The cumulative time loss (as evidenced in GDrive_1) is not merely a result of data volume, but a consequence of a workflow that forces data analysts to perform repetitive, low-level cognitive tasks.

3. **Knowledge Bottleneck:** The externalization of tacit knowledge (e.g., accounting heuristics found in GDrive_3) is currently non-existent, leaving the process vulnerable to human fatigue and scaling limitations.

These synthesized insights provide the empirical evidence required for the Practical Problem Diagnosis in Section 4.2, where the systemic causes of these issues are scrutinized. Furthermore, they serve as the direct justification for the specific functional and non-functional requirements defined in Section 4.3 .

4.2 Practical Problem Diagnosis and Root Cause Analysis

This section diagnoses the systemic inefficiencies identified during the Framework Analysis. By modeling the current manual "As-Is" workflow, this study identifies the critical cognitive bottlenecks and performs a Root Cause Analysis to justify the design requirements of the artefact and why a cognitive AI approach is necessary over traditional rule-based automation.

4.2.1 Modeling the As-Is Data Curation Workflow

The current data curation process at the case organization consists of two primary sub-processes: Data Collection and Data Preprocessing. Figure 4.1 illustrates the high-level orchestration of these activities.

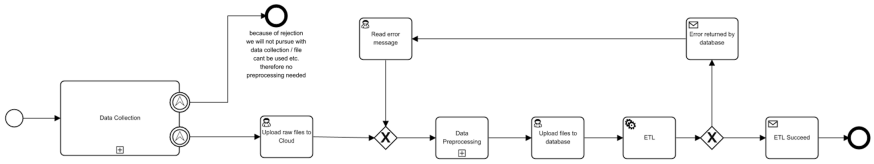


Figure 4.1: As Is Model

- **Sub-process A: Data Collection.** As modeled in Figure 4.2, this phase involves manual email correspondence with Finnish and Swedish municipalities, tracking status in Excel, and manual ingestion of raw attachments into AWS S3. While this phase is currently manual and exhibits significant potential for robotic automation (RPA), it is characterized by repetitive, administrative tasks with low cognitive complexity.

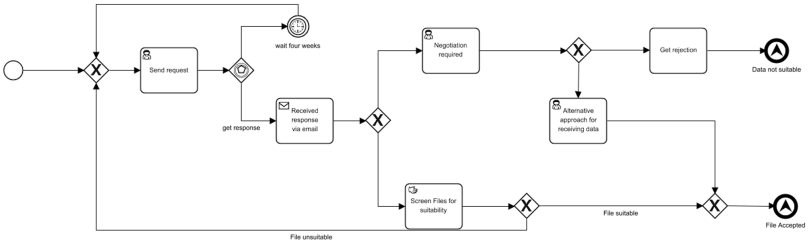


Figure 4.2: Data Collection Sub-process

- **Sub-process B: Data Preprocessing.** As modeled in Figure 4.3, this is the core focus of this study. This phase requires analysts to interpret heterogeneous purchase ledger datasets and ensure compliance with the rigid AWS Glue ETL requirements.

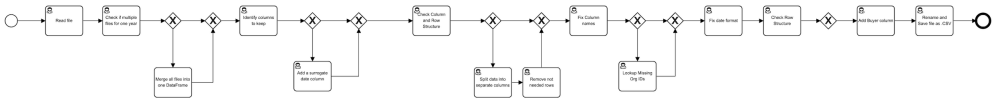


Figure 4.3: Data Preprocessing Sub-process

Although the manual nature of data collection contributes to the overall lead time, this research specifically focuses on the Data Preprocessing sub-process. This strategic focus is justified by the high concentration of tacit knowledge and cognitive load required in this stage, which represents the primary barrier to value co-creation.

4.2.2 Identification of Cognitive Pain Points

By mapping the themes from Section 4.1.2 onto the BPMN models, a critical "Cognitive Pain Point" is identified in the Preprocessing sub-process (see Figure 4.3).

As evidenced by the internal logs, the downstream ETL pipeline acts as a rigid gatekeeper. Any semantic ambiguity in the input data—such as split invoice rows or non-standard date formats—results in job failure. Consequently, analysts are forced into a cycle of manual sense-making, where they spend an estimated 60-120 hours per project (Table 4.1) performing micro-level error corrections that a deterministic script cannot handle.

4.2.3 Root Cause Analysis: The Cognitive Disconnect

To understand why this process remains heavily manual and resists traditional automation, a Root Cause Analysis (RCA) was conducted. The analysis reveals that the fundamental bottleneck is a Cognitive Disconnect between the chaotic nature of the input data and the rigid constraints of the output pipeline:

1. **Complexity Exceeding Deterministic Logic:** The core reason this process has not been automated via traditional scripts (e.g., standard Python ETL or rule-based RPA) is the inherent ambiguity of purchase ledger data. While simple formatting can be mapped to deterministic rules, resolving issues of semantic misalignment that requires probabilistic, context-aware reasoning. A purely rule-based approach would be overly brittle and fail when encountering unstructured variations.

2. **Over-reliance on Human Tacit Knowledge:** Because traditional automation cannot handle this semantic gap, the human analyst is forced to act as the "cognitive bridge." The organization heavily relies on the analysts' tacit financial heuristics (e.g., using personal judgment to balance T-accounts, as noted in GDrive_3).

3. **The Knowledge Transfer Barrier:** As long as the decision-making logic remains trapped as tacit knowledge in the minds of the analysts, the process cannot scale.

This diagnosis concludes that transitioning from the current manual state requires more than just digitizing rules; it requires transferring cognitive capabilities. Therefore, to effectively replace the manual cognitive choke points (and to outperform a baseline rule-based system), the proposed artefact must be an **AI-powered cognitive workflow** capable of externalizing tacit knowledge. The specific requirements for this artefact are defined in Section 4.3.

4.3 Defining Artefact Requirements

4.3.1 Outlining the Artefact and Eliciting Requirements

The artefact developed in this study is classified as an instantiation, as a theoretical model or construct would not adequately solve the practical problem at hand. Because instantiations are working systems used in practice, they inherently embed knowledge [1]. This study leverages this property by transferring tacit human knowledge into an AI-powered workflow solution.

From a high-level perspective, the artefact is an autonomous payment-data preprocessor capable of handling formatting tasks, basic data wrangling, and information retrieval. Its primary function is to transform raw payment data exported from various systems into a standardized format suitable for ETL operations and subsequent business intelligence applications. Based on the data collection and analysis for explicating the problem, the requirements were elicited to relate them into the Root causes, as discussed earlier.

Table 4.2: List of Requirements

REQUIREMENT ID	DESCRIPTION
REQ1	Artefact shall recognize and identify wanted columns as stated in documents Confluence_8 and Confluence_3
REQ2	Artefact shall normalize all wanted column names to snake_case
REQ3	Artefact shall append required missing columns, including buyer and a surrogate billing date column if date columns are missing..
REQ4	Artefact should be able to lookup supplier id from internal identifiers if official ID's are missing.
REQ5	Artefact shall adhere strictly to CSV upload requirements from document Confluence_3
REQ6	Artefact shall route complex or malformed files for further review.
REQ7	Artefact shall normalize data formats to strings, monetary amounts as numeric string as dates into date type.
REQ8	Artefact shall fetch datasets directly from the cloud for preprocessing purposes.
REQ9	Artefact shall resolve accounting issues as describe in document Gdrive_3
REQ10	Artefact shall upload processed files directly to the cloud for ETL operations.
REQ11	Artefact shall preprocess files autonomously when deemed possible, executing all formatting steps without requiring manual human intervention at each stage.
REQ12	Artefact shall split files chronologically by year if a dataset includes multiple years.
REQ13	Artefact shall concatenate files if a single year's data is divided across multiple files.

4.3.2 Prioritizing of Elicited Requirements

Every requirement if implemented would be valuable for the current process, however due to time constraints and the scope of the thesis, prioritization of requirements is considered not only mandatory but smart. The prioritization criteria in this study follow typical proposed criteria, which

according to Pohl [64] are cost of implementation, risk, damage due to unsuccessful implementation ,volatility and how long does it take to implement. By using these criteria the requirements were classified by using priority classes from the MoSCoW method.

Table 4.3: Requirements prioritized

PRIORITY	REQUIREMENT
`Must-Have`	REQ1-3, REQ5-7, REQ9, REQ11
`Should-Have`	REQ4, REQ8, REQ12-13
`Could-Have`	REQ10

5

The Artefact

This chapter details the architectural design, procedural logic, and technical instantiation of the proposed AI-powered data preparation workflow. Developed under the guidelines of DSR [61], the artefact operationalizes the empirical requirements identified in Chapter 4 into an automated, executable system within the n8n orchestration platform. The n8n platform was selected as the design environment due to its prominence as one of the leading workflow automation tools on the market. This automated solution integrates cognitive reasoning capabilities to manage complex semantic variations while relying on synchronized external storage.

Google Drive was chosen for this external storage synchronization, as it was already the established workspace at Tendium. Furthermore, Gemini 3.1 Lite was selected as the Large Language Model (LLM) due to its seamless integration within the Google ecosystem and its cost-effectiveness as a free-to-use tool for prototyping.

5.1 Requirements Traceability Matrix

A foundational principle of design rigor in information systems research is traceability. To ensure that the engineered artefact directly addresses the diagnosed organizational and technical problems, every architectural feature is explicitly mapped back to the formalized list of requirements.

Table 5.1 presents the comprehensive Requirements Traceability Matrix. Following the MoSCoW prioritization scheme, requirements are categorized into Must Have, Should Have, and Could Have. Crucially, to preserve empirical validity, requirements currently outside the immediate engineering scope of this version are explicitly marked as Not Implemented and left blank, establishing a transparent foundation for future iterations and organizational evaluation.

Table 5.1: Requirements-to-artefact Traceability Matrix

Requirement ID & Priority	**Status**	**n8n artefact Implementation Feature**
`REQ1` (Must Have)	Implemented	the system reads the binary file, and a custom AI node identifies the presence of mandatory columns specified in Confluence_3 and 8.
`REQ2` (Must Have)	Implemented	Executed via a deterministic formatting node that automatically parses all valid column headers and normalizes them into standard `snake_case`.
`REQ3` (Must Have)	Implemented	A programmatic appending node injects missing metadata structures (buyer identity, and normalized timestamps) before final schema validation.
`REQ4` (Should Have)	Not Implemented	
`REQ5` (Must Have)	Implemented	The final output node enforces rigid compliance with Confluence_3.
`REQ6` (Must Have)	Implemented	Controlled by a central validation Switch node. If a file is malformed or lacks mandatory columns, the automated sequence halts and immediately routes the file to the HITL manual review path.
`REQ7` (Must Have)	Implemented	Data type casting is programmatically executed using JavaScript Code nodes to strictly standardize string outputs, numeric monetary strings, and formal date types.
`REQ8` (Could Have)	Not Implemented	
`REQ9` (Must Have)	Implemented	Integrates an advanced Advanced LLM Chain node that utilizes structured prompts based on GDrive_3 to resolve complex multi-row accounting anomalies.
`REQ10` (Could Have)	Not Implemented	
`REQ11` (Must Have)	Implemented	Complete end-to-end automated orchestration across sequential n8n nodes. Once initialized, the workflow executes all standard formatting tasks autonomously without human intervention.
`REQ12` (Should Have)	Not Implemented	
`REQ13` (Should Have)	Not Implemented	

5.2 BPMN Process Redesign

5.2.1 The To-Be Workflow Model

To address the profound systemic inefficiencies and heavy cognitive load diagnosed in the manual "As-Is" workflow 4.2, the data preprocessing sub-process was fundamentally re-engineered. The redesigned "To-Be" workflow transitions the operation from a linear, human-dependent sequence into a stratified, layered architecture orchestrated by the n8n platform. Figure 5.1 illustrates the formalized To-Be process model using BPMN notation.

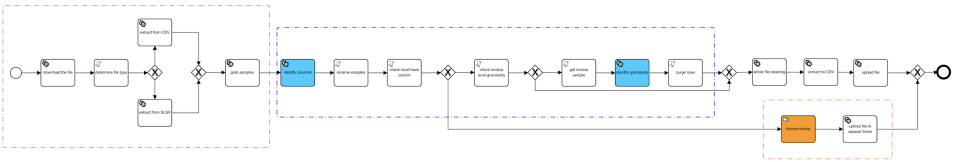


Figure 5.1: To-Be BPMN Model

As delineated in the boxed regions of Figure 5.1, the tactical innovation of the To-Be workflow lies in its modular, Three-Layer Collaborative Architecture. Each layer is visually and logically isolated based on the underlying nature of the task complexity (deterministic vs. probabilistic) and the required domain expertise:

1. **The Automated Baseline Layer (Grey Block):** This layer encapsulates the deterministic, programmatic rule-base of the system. Operating entirely without human intervention, these foundational nodes are automatically triggered upon the ingestion of new raw data packages into the synchronized Google Drive directory. This baseline layer autonomously executes secondary technical tasks, including process initialization, file downloading, character encoding validation, and primitive document format identification. By offloading these repetitive, zero-cognitive administrative steps, the system compresses the non-value-adding lead time.

2. **The Cognitive AI Reasoning Layer (Blue Block):** Positioned at the core of the process, this layer handles semantic ambiguities that traditionally resisted rule-based automation. Utilizing advanced LLM cognitive nodes, this layer processes probabilistic data transformations. Specifically, it executes contextual reasoning to identify, match,

and map heterogeneous column headers into normalized `snake_case` schemas (REQ2). Furthermore, it dynamically parses complex invoice structures, such as the multi-row transaction ledgers formats (REQ9), by executing embedded financial heuristics. This layer effectively converts the implicit domain knowledge of senior analysts into structured digital reasoning assets.

3. **The Human-in-the-Loop Gateway (Orange Block):** Operating as the ultimate data governance barrier, this layer represents the singular, strictly defined point of manual intervention within the artefact (REQ6). Designed under the principles of HITL oversight, this exception-handling loop is only activated when a structural validation node detects that a raw dataset completely lacks mandatory columns required for downstream ETL compliance. Rather than allowing corrupt data to propagate through the network or causing a hard system crash, the workflow systematically isolates the malformed file and routes it back to the human analyst for qualitative reconciliation or manual vendor follow-up.

5.2.2 The Redefinition of Human Role

The macro-level transition from the manual As-Is process to the choreographed To-Be workflow prompts a significant paradigm shift in human agency. Within the manual state, human analysts acted as the low-level processing engine, enduring immense cognitive fatigue by manually editing cells in Microsoft Excel.

In the To-Be paradigm, the analyst is elevated from a manual data manipulator to a more of a Data Steward type role. By delegating large amount of the standard formatting to the automated baseline and cognitive layers, the human expert only intervenes defensively at the orange gateway. This reduction of cognitive friction establishes the core mechanism of action for knowledge transfer, allowing human attention to focus exclusively on systemic exceptions and high-level data authenticity verification.

5.3 Artefact Design in n8n

While the "To-Be" BPMN model established in Section 5.2 provides a conceptual blueprint for process re-engineering, the actual engineering of an artefact requires adaptation to underlying technical constraints and

computational efficiency targets. This section details the concrete instantiation of the intelligent n8n process automation (IPA) workflow within the n8n orchestration platform.

Figure 5.2 illustrates the complete technical layout and node configuration of the active n8n artefact.

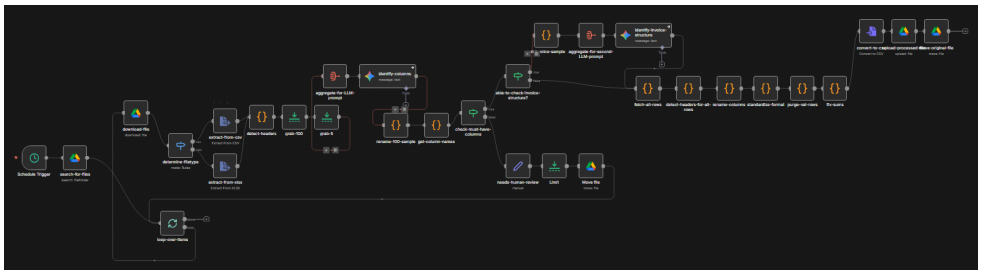


Figure 5.2: n8n full workflow

The designed n8n workflow executes the data preparation task through five distinct sequential phases, separating deterministic scripting nodes from probabilistic LLM cognitive nodes to optimize token consumption and maintain predictable data flows.

5.3.1 Phase 1: Ingestion, File Downloading, and Structural Extraction

The pipeline is initialized by a timed Trigger node that daily monitors a designated input Google Drive directory for incoming datasets. Upon detecting a new file, the workflow triggers, downloads the binary payload and utilizes programmatic extraction nodes to capture basic structural metadata. This baseline automation step isolates the initial file and establishes the operational dataset variables without human intervention, effectively suppressing the zero-value administrative lead times documented in Chapter 4.

All files ingested into the drive adhere to a standardized naming convention. The format, structured as organizationNumber_year_name, ensures that each file remains traceable to its respective organization and year. As this standardization is completed during the earlier stages of the curation workflow, it falls outside the operational scope of the artefact as it stands in its current form.

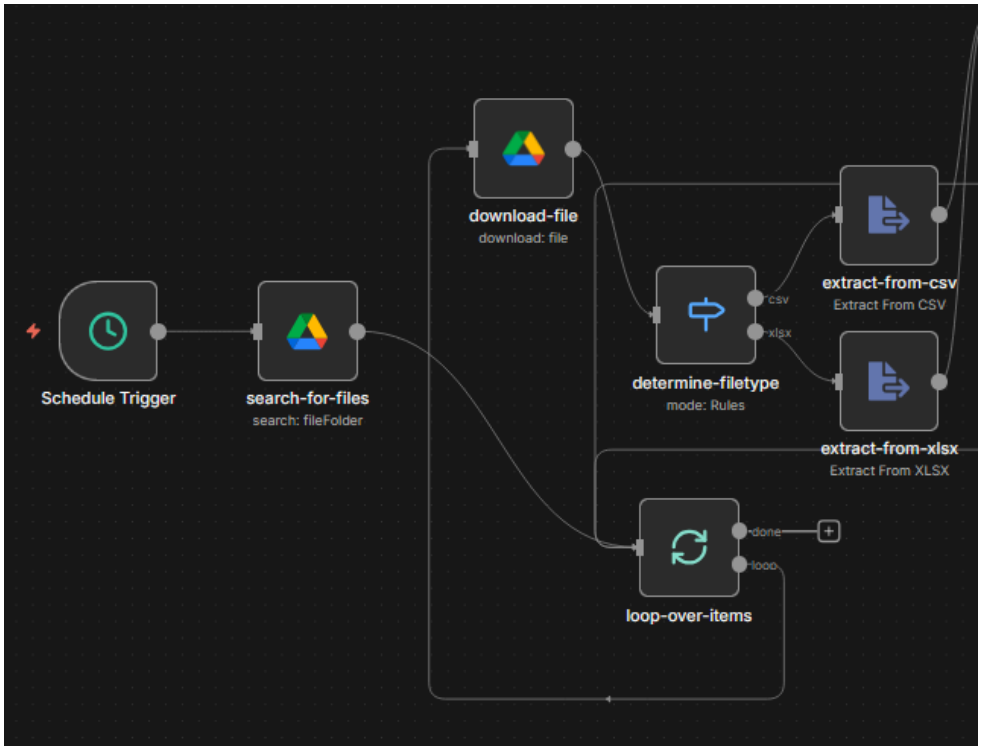


Figure 5.3: File Ingestion

The node “loop-over-items” is added to make sure the whole workflow is running once per file and so that each file is treated individually by the upcoming nodes. In practice this means that the workflow can handle e.g. 20 files but executes them one-by-one.

5.3.2 Phase 2: 5-Row Cognitive Sampling for Schema Mapping

The processed payload from Phase 1 is first routed into the initial cognitive validation module utilizing a 5-row data sampling strategy. Because evaluating column semantics does not require deep historical context, the workflow sends only a micro-sample to the first AI Node identify-columns to minimize API latency and token consumption.

This node evaluates the linguistic structure of the headers and dynamically maps heterogeneous vocabulary into the official schema. Crucially, it executes a deterministic Critical Date Rule: if all mandatory temporal columns (e.g., billing, booking) are missing, the AI autonomously infers and generates a surrogate date utilizing the fiscal year extracted from the file’s nomenclature.

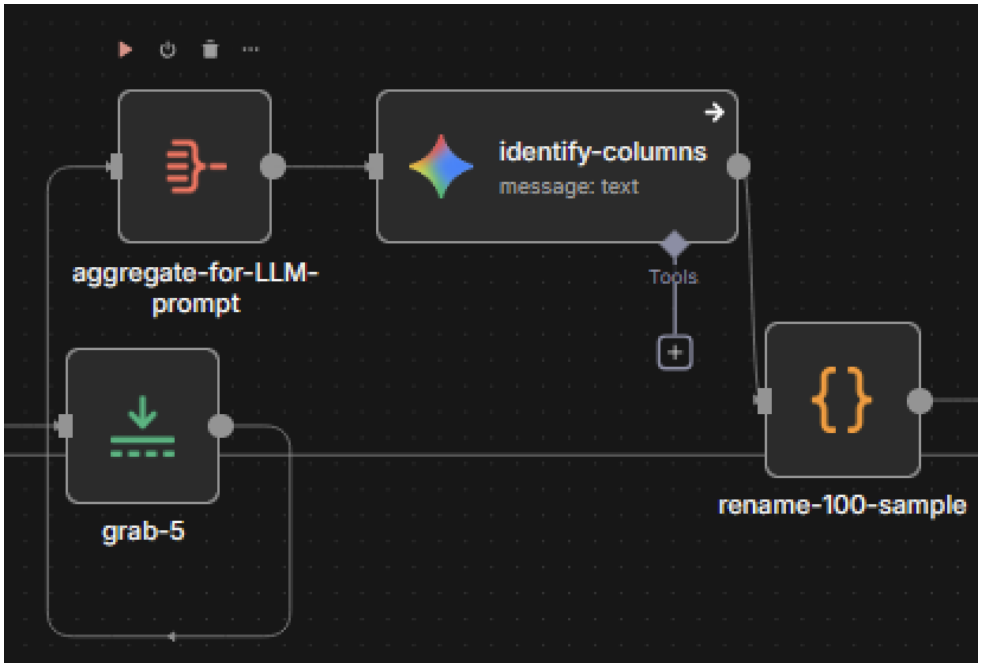


Figure 5.4: Column Renaming AI Node

5.3.3 Phase 3: Schema Application and HITL Validation Gateway

Once the 5-row sample successfully generates the mapping JSON, these structured renaming rules are applied to a larger 100-row sample of the dataset. Immediately following this schema application, the workflow introduces a critical "Fail-Fast" deterministic barrier: HITL Validation Gateway.

Before the system invests in further reasoning, a deterministic node scans the newly renamed sample for mandatory identifier columns. If absent, the automated sequence halts entirely. The corrupted file is rerouted to a designated Manual Verification Folder in Google Drive, fulfilling REQ6. This specific sequence ensures that fatal structural errors are isolated by human oversight before they can trigger downstream AI hallucinations.

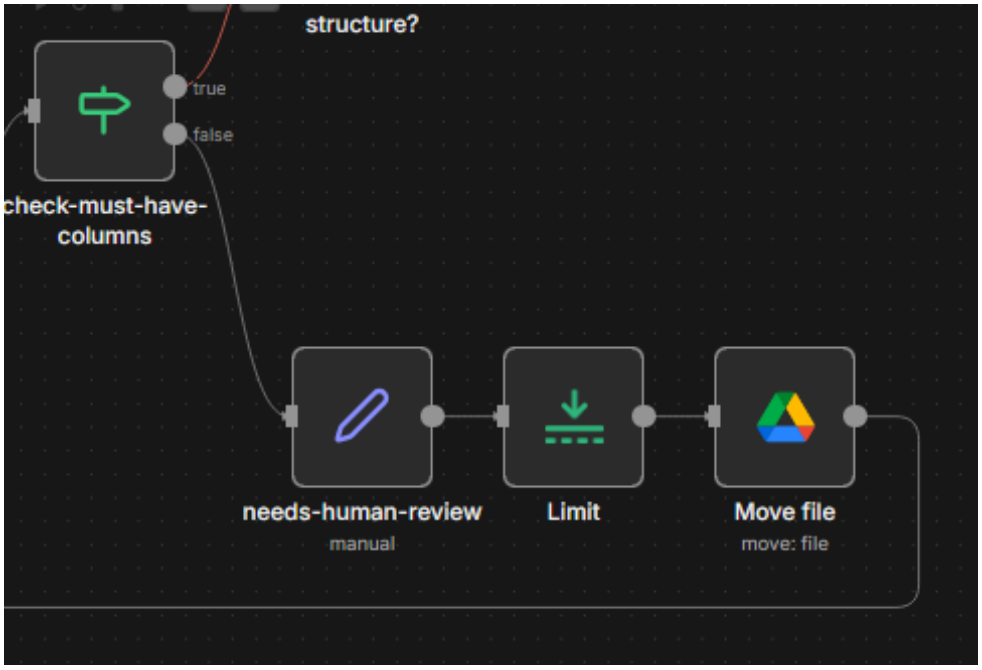


Figure 5.5: HITL Node

5.3.4 Phase 4: 100-Row Deep Sampling for Financial Anomaly Resolution

If the data set successfully passes the HITL validation gateway, the 100-row sample is forwarded to the second, more advanced AI Node (identify-invoice-structure).

To resolve complex ledger anomalies where transactions are fragmented across multiple rows, this AI node requires the expanded 100-row window to evaluate multi-row accounting relationships. Driven by strict system prompts, the LLM scans for explicit financial concepts (i.e., `has_separate_vat_row` and `has_refundable_deposit_row`) to dynamically identify and purge non-expense transaction paths. Furthermore, it applies dynamic attribute re-classification: if VAT rows are successfully identified and guided to be purged, the node enforces a rule to rename the target column from `amount_including_vat` to `amount_excluding_vat` (REQ9). By positioning this cognitive task after the HITL gateway, the artefact ensures that advanced probabilistic reasoning is only executed on structurally sound datasets. Moreover, the second IF-node after must-have-columns checks only files with necessary columns to get a sample. This includes columns for account details as well as the invoice/verification identification.

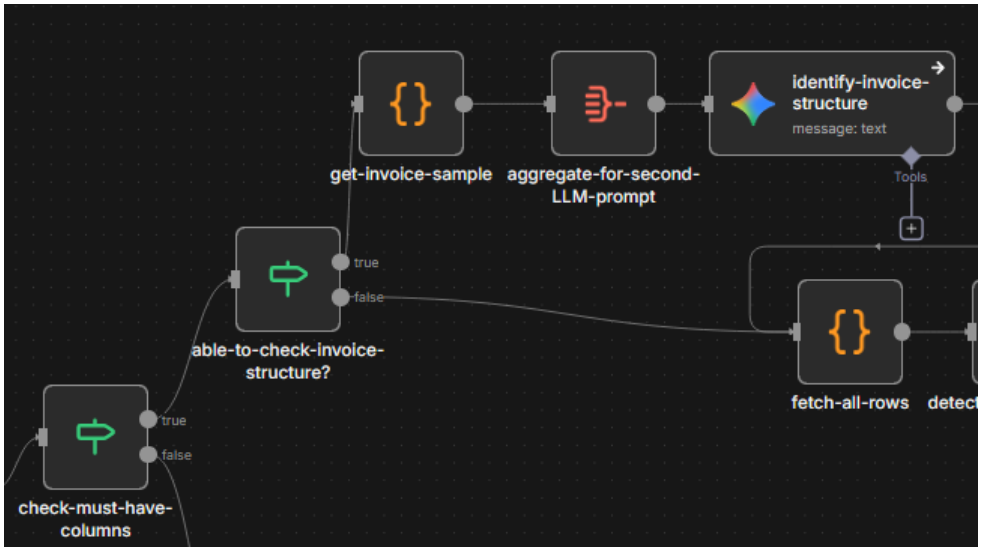


Figure 5.6: Anomaly Checking AI Node

5.3.5 Phase 5: Document Conversion and Production Export

Files that successfully pass the validation gate or bypass the HITL exception loop enter the final output module. A set of programmatic nodes executes the final document conversion, casting data types into absolute string or numeric formats and ensuring standard delimiter structures. Finally, a Google Drive Upload node automatically transfers the fully compliant, cleaned CSV file into a designated production output folder, ready to be ingested into the ETL pipeline.

5.4 Tacit Knowledge Transfer in the Design

The design process of the n8n artefact does not merely represent a technical optimization of data flows; rather, it serves as a formalized socio-technical mechanism for tacit knowledge transfer. Within the context of knowledge-intensive tasks, operating heuristics often reside either tacitly within the minds of individual domain experts or explicitly within internal work-guides. This section analyzes how the artefact design externalizes this tacit intuition into explicit digital assets across macro, micro, and defensive dimensions.

5.4.1 Workflow as an Organizational Knowledge Base

Prior to the design of this artefact, the capability to read, diagnose, and repair highly erratic municipality purchase ledgers was completely decentralized. Experienced data practitioners relied on internalized experience to interpret raw errors, rendering the knowledge transient. By defining the To-Be model within n8n, this tacit individual experience is externalized and codified into a centralized, visible, and algorithmic flow structure. The orchestrated workflow ceases to be a passive software utility; instead, it transforms into an explicit organizational knowledge repository that preserves and standardizes domain expertise.

5.4.2 Translating Heuristics into AI Nodes

The externalization of tacit knowledge is operationally executed within the two sequential cognitive sampling layers of the n8n environment. In alignment with the agency distribution principles articulated by Zhang et al. [40], this artefact does not treat AI as a monolithic black box. Instead, it systematically dissects and codifies human accounting heuristics into explicit, deterministic constraints embedded within the LLM's system prompts.

Table 5.2 illustrates the direct translation of implicit human reasoning into structured JSON generation rules within the workflow.

As demonstrated in Table 5.2, the artefact successfully translates abstract human cognition into rigid, executable logic. In the Column Renaming AI Node, linguistic intuition is captured through the Critical Date Rule. More profoundly, in the Anomalies Checking AI Node, highly specialized financial balancing logic is externalized. By providing the LLM with a strict Nordic purge vocabulary and explicit fallback conditions, the artefact bounds the probabilistic nature of the AI. It forces the LLM to output precise JSON arrays that act as deterministic triggers for downstream structural transformations.

This rigorous prompt engineering confirms that tacit organizational knowledge can be securely captured and codified as digital assets, satisfying the criteria for human-AI value co-creation at the micro-task level.

5.4.3 Defensive Knowledge Boundary and Exception Management

Finally, the HITL intervention gate serves a dual theoretical purpose. Operationally, it functions as a data governance safety mechanism that halts

Table 5.2: Tacit Knowledge Externalization: Mapping human insight to Explicit Prompt Rules

Tacit human insight (Experiential Knowledge)	**Explicit System Prompt Codification (n8n artefact)**	**Target Cognitive Node**
Temporal Metadata Inference: Analysts intuitively know that if an invoice lacks formal billing dates, a surrogate date can be logically deduced from the fiscal year denoted in the filename.	**Critical Date Rule:** "If 'billing_date' is completely missing, add a surrogate 'billing_date' utilizing the year extracted from the filename string."	Column Renaming AI Node (5-Row Sampling)
Tax and Deposit Disambiguation: Data experts visually scan for specific multi-lingual regional vocabulary (e.g., Swedish/Finnish tax terms) to identify non-cost VAT or deposits that must be excluded.	**Conceptual Purge Flags:** The prompt commands the LLM to identify abstract financial structures (`has_separate_vat_row`, `has_refundable_deposit_row`) and output the exact local keywords causing the purge.	Anomalies Checking AI Node (100-Row Sampling)
Dynamic Ledger Reclassification: If tax rows are eliminated, analysts know the remaining cost reflects the net amount, requiring a corresponding property change.	**Column Renaming Logic:** "If rows are purged because VAT is true, you MUST include a rule to rename `amount_including_vat` to `amount_excluding_vat`."	Anomalies Checking AI Node (100-Row Sampling)
Unallocated Proxy Recognition: If an invoice exhibits no explicit expense rows, accountants recognize it as a macro-level liability placeholder.	**Fallback Rule:** "If applying rules results in ALL rows being categorized as purgeable, flag `is_unallocated_liability_p` `true`."	Anomalies Checking AI Node (100-Row Sampling)

execution and redirects unsuitable files to a designated human verification folder in Google Drive.

In terms of theory, this gateway defines the absolute boundary of current explicit knowledge. When an unprecedented data failure or structural omission bypasses the cognitive layers, the file is isolated for manual human evaluation. The deliberate actions taken by the analyst to manually remediate these catastrophic exceptions create a visible data trail. This feedback loop allows researchers and data stewards to iteratively update the system prompts, thereby capturing new tacit insights and continuously expanding the artefact's explicit knowledge boundaries. This iterative loop completes the socio-technical transfer mechanism, providing the operational substrate for the human-AI value co-creation dynamics evaluated in Chapter 6.

6

Evaluation

This chapter details the ex-ante, artificial evaluation of the developed n8n artefact. Using a controlled laboratory experiment approach with real-world historical data, the evaluation investigates the artefact's operational efficacy, efficiency, and its capacity to facilitate human-AI VCC.

6.1 Evaluation Strategy and Setup

In alignment with the methodology outlined in Chapter 3, this study adopted a formative lab experiment approach. To ensure the evaluation robustly tests the artefact's generalizability and resilience against real-world data entropy, a stratified sample of 20 historical purchase ledger datasets was extracted from the organization's repository.

6.1.1 Dataset Stratification and Characteristics

Rather than artificially engineering a uniform test environment, the 20 sampled files naturally reflect the chaotic distribution of purchase ledger data. The dataset includes an even geographical split (10 from Sweden, 10 from Finland) to test cross-linguistic semantic parsing. Based on their inherent data quality, the files were categorized into three classifications:

- **Standard Files:** Datasets requiring primarily structural normalization. This includes column renaming, delimiter adjustments, and resolving files missing 'date' or 'buyer' columns, which can be deterministically inferred and added by the AI node using file nomenclature.

- **Semantic Anomaly:** Files containing complex, multi-row accounting variations or highly ambiguous variable names requiring probabilistic cognitive reasoning for resolution.
- **Missing Must Column:** Files lacking critical functional identifiers (e.g., name of the supplier, or invoice amounts with/without VAT), specifically selected to test the artefact's defensive HITL routing capability.

6.1.2 Defining Evaluation Metrics

Evaluating an AI-empowered artefact designed for KIP requires transcending traditional, one-dimensional software testing metrics (i.e., pure execution speed). Drawing upon the foundational Intelligence Augmentation (IA) framework proposed by Engelbart [65] and the Levels of Automation (LOA) model by Parasuraman et al. [66], this study establishes a multidimensional evaluation strategy.

According to Parasuraman et al. [66], automation is not an "all-or-none" construct but exists across four distinct stages: information acquisition, analysis, decision selection, and action implementation. Since the developed n8n artefact dynamically shifts the level of automation based on data complexity, executing full automation for standard formatting while enforcing a HITL boundary for critical missing data, the evaluation metrics must measure both system autonomy and human-machine synergy. Consequently, the artefact's performance across the 20 stratified datasets will be assessed using the following three targeted metrics:

1. **Metric 1: Cognitive Time Allocation (Efficiency)**
   Instead of measuring absolute processing speed, which penalizes the machine for mechanical I/O limitations on massive files, this metric isolates human cognitive effort. Inspired by Engelbart's [65] premise that technological artefacts should augment human problem-solving capacity, time expenditure is divided into two sub-metrics:
   - *System Execution Time (SET):* The duration the n8n artefact operates autonomously in the background (zero cognitive load).
   - *Active Attention Time (AAT):* The actual time the human analyst spends actively engaging with the data (e.g., manual formatting in Excel vs. reviewing HITL exceptions). The primary goal is a drastic reduction in AAT.

2. **Metric 2: Routing Accuracy (Quantitative Safety)**
   This metric evaluates the artefact's defensive "Decision Selection" capability. For highly corrupted datasets (e.g., the *Missing Must Column* group), it measures the precision of the deterministic Switch node in successfully halting the pipeline and routing the file to a HITL gateway without generating False Positives.

3. **Metric 3: Output Quality (Qualitative Efficacy)**
   This metric evaluates the final exported datasets against the organization's strict ETL ingestion schema. It assesses two dimensions of output quality: first, the deterministic presence of mandatory variables and the successful resolution of complex accounting anomalies; second, the comparative retention of non-mandatory, "nice-to-have" descriptive columns between the human baseline and the AI artefact.

By employing these three metrics, the evaluation systematically determines whether the artefact merely speeds up mechanical tasks or genuinely augments organizational intelligence through appropriate human-AI agency distribution.

6.2 Quantitative Performance Analysis

To validate the artefact's performance against the defined metrics, the 20 stratified historical datasets were processed through both the manual baseline and the n8n workflow. Table 6.1 presents a representative sample of the comparative evaluation results, highlighting the variations in processing times, formatting constraints, and routing behaviors.

Table 6.1: Representative Results of the artefact vs. Manual Baseline

ID	Data Classification	Format	Rows	Manual Time (AAT)	n8n Time (SET)	HITL Triggered?
1	Standard	CSV	48,513	07:58	01:18	N
3	Missing Must Column	CSV	23,773	06:03	06:03*	Y (Manual Fallback)
10	Semantic Anomaly	XLSX	23,601	03:03	00:39	N
14	Standard	XLSX	136,969	01:27	08:54	N
15	Standard	XLSX	1,440	01:03	00:08	N
Batch Avg	**Mixed Distribution**	**-**	**-**	**02:41**	**02:18**	**5% Activation Rate**
Batch Total	**20 Stratified Files**	**-**	**-**	**53:49**	**46:00**	**1 / 20 Files Rerouted**

**Note: For ID 3, execution halted, and SET converged to AAT due to HITL routing.*

**Note: n8n Total SET includes unattended machine processing times for large files.*

**Note: Batch Total manual time includes required manual upload processes.*

6.2.1 Metric 1: Cognitive Time Allocation and the I/O Processing Paradox

The empirical evaluation of cognitive time allocation demonstrates a profound shift in the quality and distribution of labor. To demonstrate the macro-level organizational impact of the artefact, aggregate and average time metrics across the entire sample group were computed, as shown in Table 6.1.

In the manual baseline state, processing the 20 purchase ledgers required a total cumulative time of 53 minutes and 40 seconds, yielding an Average AAT of 2 minutes and 41 seconds per file. Crucially, this human time represents uninterrupted, high-fatigue cognitive focus spent on repetitive formatting in Microsoft Excel. When the identical files were processed through the n8n artefact, the overall average processing time decreased to 2 minutes and 18 seconds per file. This achieves an absolute temporal efficiency gain of approximately 14% across the dataset portfolio.

However, viewed through the lens of IA [65], the most profound organizational impact is not merely this 14% absolute speed increase, but the complete qualitative transformation of the labor involved. For 19 out of the 20 files, direct human cognitive expenditure dropped to absolute zero. The tedious mapping and structuring tasks were entirely offloaded to unattended SET. Human intervention was strictly reserved for File ID 3, which successfully triggered the safety gateway.

Conversely, the evaluation uncovered an "I/O Processing Paradox" concerning absolute machine speed. The artefact's SET exhibited significant

statistical variance driven by row volume and format, rather than cognitive complexity. For instance, voluminous datasets like File ID 14 (136,969 rows) required 8 minutes and 54 seconds for the machine, whereas the human baseline achieved the same conversion in only 1 minute and 27 seconds. However, for smaller files like File ID 15 (1,440 rows), the n8n artefact drastically outperformed the manual baseline (8 seconds vs. 1 minute and 3 seconds). This dynamic is driven by two architectural factors:

- **Serialization Overhead (Item-per-row iteration):** While human bulk column renaming in Excel operates at an $O(1)$ temporal complexity, the n8n data orchestration processes JSON payload arrays sequentially. Execution time scales linearly with row count. Therefore, AI vastly outperforms humans on moderate files (ID 15) but encounters mechanical latency on massive datasets (ID 14).
- **Format Parsing Latency:** Parsing proprietary spreadsheet formats (`.xlsx`) introduces massive computational overhead compared to flat text files (`.csv`). File ID 1, a CSV with over 48,000 rows, was completely processed in just 1 minute and 18 seconds, highlighting the efficiency disparity in programmatic I/O handling.

Ultimately, this absolute speed deficit in large XLSX files is academically trivial. Because the machine execution requires zero human bandwidth, it successfully liberates the analyst's cognitive capacity for high-level data governance, validating the artefact's augmentation premise.

6.2.2 Metric 2: Routing Accuracy

The evaluation confirms the absolute reliability of the deterministic defensive architecture. Within the naturally stratified sample of 20 files, only one dataset (File ID 3) exhibited severe structural corruption, lacking mandatory functional identifiers such as supplier names and monetary amounts.

The artefact's validation Switch node successfully detected this omission, achieving a 100% True Positive interception rate. The automated sequence was immediately halted, and the file was accurately routed to the manual review pool, causing the workflow time to converge with the manual AAT (06:03). Furthermore, none of the 19 viable files were falsely redirected, resulting in a 0% False Positive rate. This flawless routing confirms that the artefact safely enforces the defined knowledge boundary (`REQ6`).

6.3 Output Quality Analysis

While Metric 2 evaluates input interception, Metric 3 evaluates the final generative output. For the datasets that successfully pass the initial gateway, this metric assesses whether the AI cognitive nodes reliably mapped and retained all mandatory variables in the final exported files without generative loss. Furthermore, it evaluates the successful resolution of complex accounting anomalies and the comparative retention of non-mandatory, "nice-to-have" descriptive columns.

6.3.1 Mandatory Compliance and Anomaly Resolution Efficacy

As illustrated in Figure 6.1, the files were evaluated in their output in having must-have columns for the data to be considered sound, fitness to overall ETL ingestion schema as well as how well they would deal with accounting structures, described in chapter 4.

File ID	Output has must-have columns		Fits ETL schema		Delt with accounting anomalies	
	Manual	n8n	Manual	n8n	Manual	n8n
1	✓	✓	✓	✓		
2	✓	✓	✓	✓		
3	–	–	–	–		
4	✓	✓	✓	✓	✓	✓
5	✓	✓	✓	✓		
6	✓	✓	✓	✓		
7	✓	✓	✓	✓		
8	✓	✓	✓	✓		
9	✓	✓	✓	✓		
10	✓	✓	✓	✓	✓	✓
11	✓	✓	✓	✓		
12	✓	✓	✓	✓		
13	✓	✓	✓	✓	✓	✗
14	✓	✓	✓	✓		
15	✓	✓	✓	✓		
16	✓	✓	✓	✓		
17	✓	✓	✓	✓		
18	✓	✓	✓	✓		
19	✓	✓	✓	✓	✓	✓
20	✓	✓	✓	✓	✓	✓

Figure 6.1: Output Quality Evaluation Matrix across 20 Cleaned Files

Excluding File ID 3, which was deliberately halted by the HITL safety gateway, the n8n artefact demonstrated solid baseline reliability. For the remaining 19 processed files, the human analyst and the AI workflow achieved a 100% success rate in extracting the necessary "must-have" columns and formatting the outputs to fit the rigid ETL ingestion schema.

However, a critical divergence emerged when evaluating the resolution of Semantic Anomalies. Within the sample, exactly five documents contained complex, multi-row accounting variations that required deep contextual reasoning to flatten into single-row operational expenses. The manual human baseline achieved a 100% success rate across these five files.

The n8n artefact successfully resolved these anomalies in four out of the five instances (an 80% success rate). It successfully purged VAT rows and dynamically renamed attributes. The singular cognitive failure occurred in File ID 13. In this particular case, information regarding the VAT rows was present in another column that was neither the number of the transaction account name of the transaction account, making it an edge-case that the current design was not ready to handle.

This singular peculiar failure highlights a limitation of bound probabilistic models. Whilst they excel at executing predefined heuristic patterns, they lack the human analyst's fluid adaptability when confronting entirely unprecedented structural anomalies.

6.3.2 Selective Attention and Differences in Non-Mandatory Variables

While both the human baseline and the artefact performed robustly on mandatory ETL requirements, an analysis of the column mapping outputs reveals behavioral discrepancies regarding non-mandatory, "nice-to-have" variables. In the 19 files analyzed, the human workflow and the artefact agreed on a total of 142 values. This means that for 142 separate column key-value pairings—where the "key" refers to the original column header and the "value" refers to the accepted column name—the human manual workflow and the artefact aligned. In total, the human extracted 151 columns and the artefact extracted 155 columns, with 147 columns shared between both. This results in an overall agreement rate of 96.6%. The table below illustrates these findings.

Table 6.2: Mapping Results of Columns

FILE ID	Human	AI-node	Shared	Value Agree	Value Differ	Result
1	8	8	8	8	0	Perfect
2	8	7	7	7	0	AI missed
4	9	9	9	8	1	Value diff
5	9	9	9	9	0	Perfect
6	6	8	6	4	2	Difference between two mappings
7	10	10	10	10	0	Perfect
8	6	7	6	6	0	AI added Extra
9	11	10	10	10	0	AI missed
10	7	8	7	7	0	AI added Extra
11	9	9	9	9	0	Perfect
12	9	9	9	9	0	Perfect
13	8	9	8	6	2	AI added extra column and difference between mappings
14	7	7	7	7	0	Perfect
15	5	5	5	5	0	Perfect
16	9	9	7	7	0	Difference between two mappings
17	7	8	7	7	0	AI added extra column, difference between mappings
18	4	4	4	4	0	Perfect
19	9	9	9	9	0	Perfect
20	10	10	10	10	0	Perfect

The artefact added 8 columns outside the shared columns and missed 4. In file ID 16, the outcome differed for amounts, as the original file included four headers for the invoice amount. The key difference between these four headers was that two were invoice amounts and two were referred to as “redovisad” (reported amount). The human analyst kept the reported amounts, whereas the AI-node kept the invoice amounts. In file IDs 2 and 9, the AI clearly missed columns referring to the invoice’s cost center, which the human analyst successfully identified.

In other scenarios, the AI-node added extra columns that the human missed. These included a verification number for file ID 10, a VAT amount and verification number for file ID 6, and a cost center for file ID 13; all of these were correct mappings. However, in two scenarios, the AI extracted the wrong column which the human had intentionally omitted. In file ID

8, this included a column for the invoice deadline date, which the AI incorrectly mapped to the invoice payment date. In file ID 17, the artefact mapped the original header "uppl.datum" as the invoice booking date, whereas the human left this column out.

Disagreements regarding original header values also revealed qualitative differences. In file ID 6, a column simply labeled "date" ("päiväys" in Finnish) was mapped as a booking date by the AI, while the human mapped it as a billing date. A second disagreement occurred when determining whether the mapped amount included VAT; the human chose "with VAT," whereas the AI chose "without VAT." In file ID 13, the AI correctly identified "ins.datum" as the booking date, whereas the human incorrectly mapped it as the billing date. The abbreviation "ins.datum" stands for "Inskannings-datum" or "Inkomstdatum" (date of receipt in English). From the municipality's perspective, this refers to when the invoice was registered, not when the supplier issued it. Another human error was captured in this same file: the human had mapped and extracted the verification number column using its original header, which would have failed to be ingest into the ETL process.

The evaluation reveals an overwhelmingly high agreement rate of 96.6% between the current manual human workflow and the designed artefact, demonstrating that both perform exceptionally well on header identification and extraction. However, an analysis of the remaining 3.4% discrepancy indicates that neither the human analyst nor the AI achieve perfect fidelity in retaining secondary, "nice-to-have" columns. In these edge cases, both exhibited different forms of "selective attention" and omission, driven by fundamentally different underlying mechanisms.

When processing extensive datasets, especially one after another for a longer period of time, human analysts often adopt a task-focused strategy, prioritizing the mandatory requirements as method of satisfying the low-hanging fruit in order required to successfully process the file. Driven by cognitive load, the human analyst actively or passively ignored several valid, "nice-to-have" columns.

When it comes to the artefact, the omissions were dictated by system prompts and probabilistic evaluations. While the AI operates within the strict bounds of its defined instructions, its decision-making relies on confidence scoring. If a secondary column header featured an ambiguous linguistic structure that did not clearly align with the defined mapping rules, the model's confidence fell below the necessary threshold, causing it to either discard or misclassify the column.

7

Discussion

This chapter reflects on the empirical and theoretical implications of the research findings. By aligning the controlled laboratory evaluation results from Chapter 6 with the extended background established in Chapter 3, this discussion elucidates the value and wider contributions of the developed artefact, while critically assessing its systemic limitations and ethical dimensions.

7.1 Key Findings

This section presents the core empirical and theoretical findings of this study by aligning them directly with the primary and sub-research questions (RQs) posed in Chapter 1. By analyzing the data patterns compiled during the laboratory experiment, the discussion unpacks how the developed artefact navigates the intersection of AI-powered automation, knowledge transfer, and human-machine synergy.

7.1.1 Codifying Tacit Knowledge into Cognitive Nodes

The first major finding directly addresses sub-RQ1 by shifting the focus from rigid programming to probabilistic reasoning. Specific data cleaning tasks for determined outputs cannot be curated conventionally using conventional deterministic scripts like Python or standard RPA. For instance, every new column name variation of a supplier name information would have to be added to a rule-based solution.

To resolve this, the artefact shows that data specialists' tacit knowledge can be effectively externalized by translating implicit human insight into explicit, structured prompts given to a LLM. The empirical execution of the artefact demonstrates that this externalization is achieved across two distinct cognitive dimensions:

- Linguistic Intuition Extraction: Through the 5-row sampling node, the analyst's experiential ability to instantly recognize misspelled or mismatched column headers under explicit invoice IDs is transferred to the machine, enabling real-time semantic schema normalization without human intervention.
- Financial Balancing Transfer: Through the 100-row deep sampling node, the complex accounting heuristics required to balance fragmented municipality transactions are successfully codified. The node applies specialized prompt boundaries to autonomously identify VAT distributions and dynamically alter column qualifiers.

Consequently, the design process reveals that tacit knowledge is not successfully integrated by duplicating every conceivable rule, but by creating modular cognitive nodes that mimic human contextual sense-making.

7.1.2 Multi-Dimensional Micro-Level Value Co-Creation

Beyond its structural integration, this study significantly advances the theoretical boundaries of VCC. As critically reviewed in Section 2.3.3, the prevailing information systems literature predominantly anchors VCC within macro-level service ecosystems [57] or customer-facing, front-end digital touchpoints [58]. This pervasive bias has led researchers to overlook internal operations, mistakenly assuming that data processing pipelines are merely mechanical backend routines ripe for blind, full machine substitution [59].

This research directly refutes this baseline assumption by operationalizing human-AI collaboration within the core business operations of a data service provider. For a data aggregator whose primary commercial product is value-added procurement insights, the purchase ledger data curation pipeline constitutes the core production line. By analyzing the laboratory experiment results, this study conceptualizes that human-AI VCC within this business operations domain manifests across three specific, interdependent value dimensions, thereby providing a robust answer to sub-RQ2:

- **Dimension 1: Cognitive Offloading and Qualitative Labor Shift**
  The evaluation demonstrated an absolute temporal efficiency gain of approximately 14% (reducing the average processing time from 02:41 to 02:18). However, co-created value in this dimension is not defined merely by this mathematical velocity, but by the liberation of intellectual capital. Co-created value in this dimension is not merely defined by temporal velocity, but by the liberation of human intellectual capital. By offloading highly repetitive, high-fatigue Excel formatting to the machine's unattended SET, the workflow elevates the human data specialist from a data manipulator to a micro-level data governor, allowing the reallocation of active human focus toward other job tasks and create more working values.

- **Dimension 2: Organizational Knowledge Retention**
  A paramount strategic dimension of co-created value identified in this study is the formalization and retention of organizational knowledge assets. Historically, the capacity to decipher multilingual municipality ledgers resided strictly as transient knowledge within analysts' minds. If a domain expert departed the firm, this operating capability was permanently lost.

  By designing cognitive AI nodes that embody these heuristics through specialized prompts, the artefact successfully captures and stores this transient human insight. The co-creation process transforms individual expertise into a permanent organizational asset, enabling the company to scale across new jurisdictions without linearly increasing expert headcount.

- **Dimension 3: Risk Mitigation and Quality Governance**
  Value is robustly co-created through the mutual mitigation of human and machine weaknesses. As demonstrated in Section 6.3, the human and AI achieved a 96.6% agreement rate on schema mapping. Their 3.4% discrepancy proved highly complementary: the AI's deterministic consistency mitigates human "tunnel vision" and cognitive fatigue, while the human-engineered HITL gateway achieved a 100% interception rate for structurally corrupt files (File ID 3). Human intelligence and machine detection thus co-create a defensive ecosystem, securing the downstream ETL pipeline.

Consequently, by mapping these three granular dimensions, this study provides empirical micro-foundations for the typology of Braathen and Findsrud [60] and the agency distribution principles of Zhang et al. [40], demonstrating that core data product engineering is optimized not through labor displacement, but through IA [37].

7.1.3 Orchestrated Augmentation Over Blind Substitution

Synthesizing the insights from the previous sub-sections, the resolution to the Main RQ manifests as an architectural shift from blind machine replacement to orchestrated augmentation.

The evaluation exposed two critical vulnerabilities of automated systems that make full human substitution untenable. First is the "I/O processing paradox": for voluminous datasets like File ID 14, programmatic serialization caused the machine to lag significantly behind human execution speeds. Second, and more importantly, is the limitation of probabilistic boundaries. As evidenced by the singular anomaly failure in File ID 13, where VAT information was embedded in an unprecedented structural format, the LLM could not adapt beyond the lexical bounds of its prompted heuristic patterns.

More importantly, the evaluation exposed the fundamental vulnerability of unguided automation when confronting chaotic data environments. This empirical behavior strongly validates the typology proposed by Braathen and Findsrud [60], which posits that for highly volatile and unstandardized data domains like municipal AP ledgers, a blunt automation model is bound to fail. When data quality issues are systemic, a rule-based algorithm cannot adapt, resulting in cascading downstream ETL crashes. This finding demonstrates that the historical institutional pursuit of "blind machine replacement" [59] is an inadequate solution for KIP.

Therefore, to facilitate sustainable knowledge transfer and value co-creation, workflows must be designed with an asymmetric distribution of agency [40]. By enforcing a Human-in-the-Loop gateway to govern edge cases, the artefact circumvents the brittleness of AI, proving that the future of data curation lies in IA rather than linear labor substitution [37].

7.2 Novelty, Theoretical, and Practical Significance

The conceptualization, implementation, and empirical validation of the developed n8n artefact contribute significantly to the information systems literature by directly addressing the intersectional research gap identified

in Subsection 2.3.4. The theoretical novelty of this research lies in its systemic integration of isolated domains and its operationalization of VCC within the core production line of data-driven organization.

7.2.1 The Integration of Historically Isolated Domains

The primary theoretical novelty of this artefact resides in its architectural capacity to integrate three socio-technical pillars—cognitive AI automation, HITL oversight, and tacit knowledge transfer, which have evolved in isolation within current IS literature. As reviewed in Chapter 2, prior automated data curation research typically suffers from a binary divide: studies either pursue brittle, full-automation models that fail under real-world data entropy [16, 60], or they conceptualize human oversight as a low-level operational bottleneck that restricts organizational scalability [59, 37].

By introducing a Three-Layer Architecture, this study presents a unified design pattern demonstrating that these domains are not structurally adversarial, but complementary. The artefact establishes a clear topological flow where probabilistic LLM reasoning nodes are safely bounded by deterministic programmatic rules at the ingestion phase, and strictly governed by a defensive validation gateway at the export phase.

This research contributes to DSR by providing a concrete architectural blueprint showing how highly ambiguous, knowledge-embedded datasets can be systematically processed without sacrificing technological agency or human governance, thereby breaking the traditional scholarly silo between automation and control.

7.2.2 VCC and Knowledge Foundations of Data Services

Beyond structural integration, this study achieves a theoretical advancement in the conceptual boundary of VCC. As critically examined in Section 2.3.3, the overwhelming consensus in existing applied research anchors VCC within macro-level service ecosystems [57] or customer-facing, front-end conversational interfaces [58]. This analytical bias has led researchers to systematically overlook internal data workflows, mistakenly treating data preprocessing as a mechanistic background routine void of collaborative value creation [59].

This research refutes this assumption by operationalizing human-AI VCC within the core business operations of an organization, in the meantime

also a data aggregator. For an organization whose primary commercial offerings are value-added procurement insights, the purchase ledger data curation pipeline is not an administrative overhead, it represents the primary core production line. By framing this operational environment through the lens of IA [37], this study proves that value is co-created across specific micro-task dimensions that carry strategic weight:

First, this research expands VCC theory by identifying Organizational Knowledge Retention as a dimension of co-created value. In service-dominant logic, human expertise acts as an operant resource, which is traditionally vulnerable to individual attrition. The process designed in this artefact successfully converts these human cognitive heuristics into explicit, stored digital assets. By capturing this intuitive domain knowledge and embedding it into the active workflow, the artefact co-creates long-term institutional resilience, retaining cognitive capability within the system and resolving the organization's dependency on individual specialists.

Second, the study appled the collaborative trade-offs of VCC by substantiating the concepts of Cognitive Offloading. The 14% contraction in AAT demonstrated in Chapter 6 proves that co-created value is not characterized by a simple race for speed, but by a shift in temporal quality. By offloading high-fatigue semantic mapping to the machine's SET, human cognitive bandwidth is liberated for higher-order market strategy. Concurrently, the HITL gateway provides empirical proof for the agency distribution principles of Zhang et al. [40] and the task-complexity framework of Braathen and Findsrud [60].

Regarding practical significance to other potential commercial data aggregators, these three dimensions of value hold true beyond Tendium for two main reasons.

First, we are addressing a universal Problem. Tendium is simply an instantiation of a much broader issue: the KIP of curating highly unstructured, ambiguous data relying on human tacit knowledge. Whether it is a bank reconciling fragmented transactions, a healthcare provider merging patient records, or another data aggregator, the fundamental friction is the same. Therefore, the mechanisms of offloading cognitive fatigue and mitigating human error are applicable to this class of organizational problems.

Second, our value dimensions are anchored in established macro-theories. Including the knowledge transfer theory and the Intelligence augmentation framework. This study is a single-case study, while what it provides not a mathematical formula, but a proven micro-foundation and an architectural blueprint. While the specific n8n nodes or data related prompts might

be unique to Tendium, the overarching design principles and the resulting dimensions of value co-creation are transferable to any organization looking to shift from blind automation to human-AI intelligence augmentation.

7.2.3 Scalability and Cognitive Offloading

From a practical perspective, this artefact introduces critical operational benefits for the organization. The empirical insights derived from the artefact's evaluation offer two profound practical implications for core business operations:

First, the artefact establishes a clear framework for low-cost scalability across multi-jurisdictional markets. Traditionally, when a data service provider expands its operations to ingest purchase ledgers from new international markets, the organization faces a steep financial and operational barrier. It must recruit or allocate specialized regional analysts to manually map local formatting variations and decipher localized accounting terminology. Current design overcomes this human resource bottleneck, giving the potential to expand its data aggregation portfolio into new geographic regions without a linear, unsustainable increase in expert headcount.

Second, the cognitive offloading represents the complete elimination of low-level, high-fatigue tasks. In the manual baseline, analysts waste valuable cognitive bandwidth executing bulk renames or tracking column typos. By offloading these repetitive administrative tasks, the workflow liberates the company's human capital. Managers can reallocate this newly emancipated intellectual capacity toward high-value, revenue-generating activities—such as advanced predictive analytics, deep data-driven client consulting, and the cross-verification of the critically flawed datasets safely isolated by the HITL defensive gateway.

7.3 Limitations of the Study

In terms of methodological limitations, the artefact was tested using only 20 datasets: 10 from Finland and 10 from Sweden. A primary issue with this sampling is the small size, meaning the evaluation likely did not capture all the edge cases that a larger, more diverse dataset would have revealed. To combat this, a Human-in-the-Loop (HITL) node was added to the artefact to catch these edge cases, making it possible to successfully run the experiment. However, more examples of non-fitting datasets should have been included to make the artefact more robust to varying format

types. Because the artefact is a prototype, future development should include automated solutions for these edge cases to further improve its output.

Second, because this study utilized Action Research (AR) as a strategy, involving a practitioner in the design process helped integrate valuable user feedback regarding usability. However, relying on a single practitioner and lacking third-party feedback is a limitation; a wider spectrum of input would have been highly beneficial for refining the artefact.

In terms of generalizability, the artefact was developed for a specific company with a specific purpose. Although data cleaning is a universal task for data specialists across organizations, the specific intent behind the cleaning and the desired output often vary. The design of this artefact, along with its identified problems and requirements, explicitly reflects its target company's needs. Because these elements are highly context-specific—as is typical for company-sponsored thesis projects—the generalizability of the artefact is limited.

Due to scope and time constraints, the artefact was restricted to handling only structured data formats, such as .csv or .xlsx. While some purchase ledger files exist in .pdf or other formats, processing these was left outside the scope of the current project. This is a notable limitation, as users must still manually format and clean these files using other tools and technologies. Implementing a PDF extractor presents a clear opportunity for future research.

What's more, as revealed by the cognitive failure in File ID 13, the artefact's semantic reasoning is bounded by the static keyword dictionary hard-coded into its prompts. When encountering entirely unprecedented regional abbreviations or structural anomalies outside this defined lexical boundary, the LLM exhibits "selective omission." This indicates that the system is not entirely autonomous, it requires a continuous feedback loop where human data stewards regularly update the prompt logic to capture evolving tacit knowledge.

Currently, the artefact cannot recognize if a single dataset from the same year is divided across multiple files, an issue identified during the data collection and analysis phase. Addressing this was elicited as a "could-have" requirement for the artefact but was ultimately not implemented. Consequently, the data still requires manual preparation, such as combining quarterly reports into a single file to process a full year's worth of data at once. As noted in confluence_17, successful data ingestion requires strictly one file per year, per buyer.

7.4 Ethical and Social Aspects

The introduction of cognitively empowered workflows into core data production lines invites critical ethical reflections, primarily concerning labor dynamics and data governance. First, from a socio-technical perspective, the artefact directly addresses the ethical anxiety surrounding technology-driven labor displacement [59]. By embedding a defensive HITL validation gateway and adhering to an IA philosophy [37], the system does not aim for full machine substitution. Instead, it upgrades the human worker's role, liberating them from high-fatigue manual Excel manipulation and repositioning them as high-level data governors and exception auditors.

Second, operational reliance on external commercial LLM APIs introduces critical data governance considerations. Although the municipal purchase ledger datasets processed by data aggregators are legally classified as public procurement records, routing large-scale corporate ingestion streams through external cloud endpoints poses potential exposure risks. Ethical data practice necessitates the implementation of strict data auditing or masking guardrails within the initial deterministic preprocessing layer. This ensures that no confidential metadata or proprietary business patterns are inadvertently leaked, balancing technical agility with corporate information security.

7.5 Suggestions for Future Research

Based on the laboratory evaluation constraints and the engineering requirements left unfulfilled in this iteration, future research trajectories should pursue two primary avenues. First, technical development should focus on expanding the structural flexibility of the artefact to achieve full pipeline end-to-end automation. Future iterations must engineer programmatic nodes capable of handling macro-level file partitioning, specifically focusing on autonomously splitting large datasets chronologically by year (`REQ12`) and concatenating fragmented seasonal files into unified relational tables (`REQ13`). Furthermore, other aspects of larger data curation workflows should be investigated.

Second, future research should investigate architectural cost optimization and localized data sovereignty. While universal LLMs demonstrated exceptional efficacy in semantic disambiguation, their proprietary, token-based transaction costs introduce long-term operational scaling constraints

for commercial data aggregators. Future studies should explore the deployment and fine-tuning of localized, open-source small language models, such as Llama-3, trained on domain-specific procurement and accounting datasets. This shift would drastically compress corporate API expenditures while entirely neutralizing external privacy risks by processing sensitive core business data within an on-premise, fully self-hosted infrastructure.

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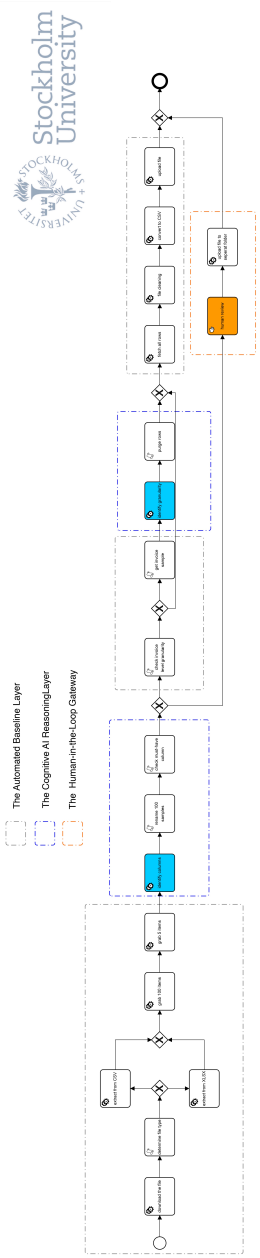
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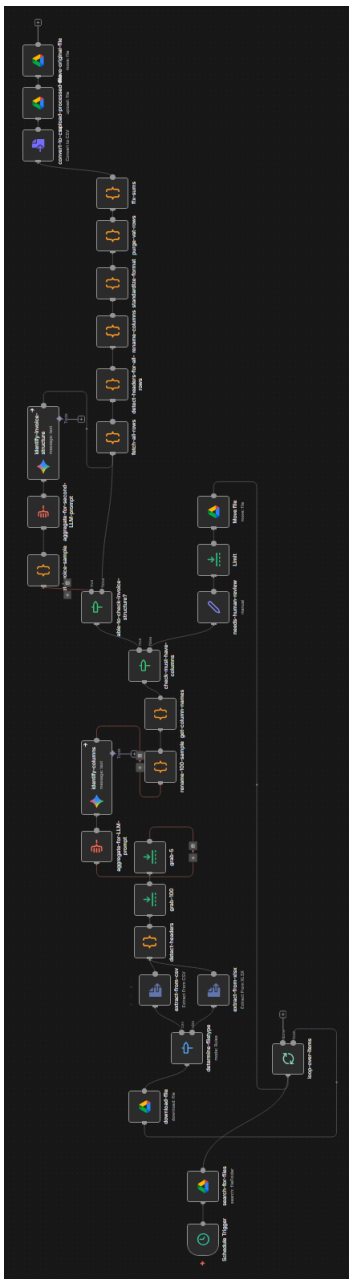
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Appendix A — To-Be BPMN Model



Appendix B — n8n Full Workflow



Appendix C — Column Identifying AI Node Prompt

Prompt: Column Identifying AI Node

```
You are an expert data analyst.  I am providing a sample
dataset containing 5 rows in JSON format.  Your task is to
map the current column headers (the keys in the JSON) to my
official, required column names.

Use the current header names and the data values within those
columns to infer the correct mapping.

Here is my list of official column names you must choose from:
- supplier_name - supplier_orgnr - amount_including_vat -
amount_excluding_vat - vat - billing_date - booking_date -
payment_date - period - invoice_number - verification_number -
account - account_name - cost_centre - cost_centre_name

CONTEXT FOR LOGIC: - Current Filename:  {{ $json.filename }}

MAPPING RULES:
1.  If a current header matches the meaning of an official
column name, map it to that official name.
2.  If a current header DOES NOT map to any of the official
names, map it to its exact original name.
3.  EVERY original header from the dataset must be present as a
key in your output.
4.  There can only be one official column mapping e.g.  if
there are two columns for accounts only one of them can be
mapped to the official column.
5.  CRITICAL DATE RULE: Look at the official column names
you have chosen for your mapping.  Did you map ANY column to
‘‘billing_date’’, ‘‘booking_date’’, ‘‘payment_date’’, or ‘‘period’’?
If YES: You are done with dates.  DO NOT add a surrogate date.
If NO (all four are completely missing):  Add a new key
‘‘billing_date’’ with the value YYYY-01-01 (use the year
extracted from the filename, e.g.  2024-01-01).
if a column is called e.g.  Bokfdag but the values inside do
not look like a date e.g.  its only a day you can leave this
column out.
6.  CRITICAL SUPPLIER & ORGNR BOUNDARY RULE:
- Identify the Supplier Name:  Look for columns containing
varying company names and map that header to ‘‘supplier_name’’.
```

```
- Dynamic vs. Static ID Rule (e.g., ‘‘Y-tunnus’’): Before
mapping an organization ID column like ‘‘Y-tunnus’’, you MUST
inspect the values within that column:
- IF the values VARY across different rows: This represents
different suppliers. Map this column to ‘‘supplier_orgnr’’.
- IF the values are IDENTICAL across all rows: This represents
the buyer (since the file is from the buyer’s perspective). DO
NOT map it to ‘‘supplier_orgnr’’. Instead, map it to its exact
original name per Rule 2.
7. ACCOUNT RULE:
- If there are multiple columns for account, keep JHS-tili for
Finland if that exists. Example: Tili, Tilin nimi, JHS-tili,
JHS-tilin nimi. In this case, keep JHS accounts.
8. DEBIT / CREDIT RULE:
- If the dataset contains separate columns for Debit and Credit
(e.g., ‘‘Debet’’ and ‘‘Kredit’’), use ONLY the Debit (‘‘Debet’’)
column for your official amount mapping (e.g., map it to
‘‘amount_excluding_vat’’ or ‘‘amount_including_vat’’).
9. ANOTHER AMOUNT RULE:
- If there are columns for total invoice amount and also row
level amount, keep the row level column as the amount column.

Here is the sample dataset: {{ JSON.stringify($json.rows) }}

OUTPUT INSTRUCTIONS:
Return ONLY a raw JSON dictionary where the keys are the
original headers (or the new required headers if rules 4 or
5 apply), and the values are the mapped official names or
fallback values. Do not include markdown formatting, backticks
(‘‘‘), or conversational text.

Example format (assuming filename is ‘‘helsinki_2024.xlsx’’
and dates/buyer were missing): { ‘‘Current_Messy_Header’’:
‘‘amount_including_vat’’, ‘‘billing_date’’: ‘‘01/01/2024’’, }
```

Appendix D — Invoice Anomaly Checking AI Node Prompt

Prompt: Invoice Anomaly Checking AI Node

```
You are an expert forensic data analyst and accountant
analyzing Accounts Payable data.
I am providing you with a JSON array containing a sample
dataset of MULTIPLE rows that share the SAME invoice number.

Your task is to investigate the structure of this specific
multi-row invoice and generate universal filtering RULES that
will be applied to thousands of similar invoices.

ANALYSIS INSTRUCTIONS:
1.  Analyze the data values, account numbers, and account
names/descriptions.  Explain WHY the invoice was split across
multiple lines (e.g., separating core expenses, VAT, or
refundable deposits).
2.  Identify which row(s) represent the actual core,
profit-impacting expense.
3.  RULE GENERATION (The Purge List):
Identify which rows represent taxes/VAT deductibles, refundable
deposits, AND the overall Accounts Payable liability row.
These must be purged to avoid double counting.
- Hint for identifying:  Look out for tax terms ('moms', 'alv',
'vat'), refundable terms ('pant', 'deposition', 'retur',
'vakuus', 'saamiset'), and liability terms ('ostovelat',
'ostovelka', 'leverantörsskulder', 'accounts payable',
'leverantörsskuld').  Examples of terms you might see but do
not assume they exist, Only extract words that ACTUALLY exist
to the provided JSON.
- PRIMARY RULE: Extract the EXACT account numbers associated
with these purgeable rows.
- SECONDARY RULE: Extract the exact keywords from the account
name/description that uniquely identify these purgeable rows.
4.  COLUMN RENAMING LOGIC:
Inspect the keys (column names) in the provided JSON. If
''amount_including_vat'' is the ONLY amount column provided
AND you determined that rows need to be purged (because
has_separate_vat_row is true OR has_refundable_deposit_row
is true), the remaining core expense row represents the
net cost for the operational period.  In this case, you
```

```
MUST include a rule to rename ‘‘amount_including_vat’’ to
‘‘amount_excluding_vat’’.
5.  FALLBACK RULE (CRITICAL):
If applying your rules from step 3 would result in ALL rows
being categorized as purgeable (e.g., the JSON only contains a
liability/Accounts Payable row and no expense rows), you must
flag ‘‘is_unallocated_liability_proxy’’:  true.

Here are the rows for the invoice:  {{
JSON.stringify($input.all().map(item => item.json)) }}

OUTPUT INSTRUCTIONS:
Return ONLY a raw JSON dictionary with your analysis.  Do not
include markdown formatting or backticks (‘‘‘).

REQUIRED JSON FORMAT:
{ ‘‘structure_explanation’’:  ‘‘Explain your findings
in detail.’’, ‘‘has_separate_vat_row’’:  true/false,
‘‘has_refundable_deposit_row’’:  true/false,
‘‘is_unallocated_liability_proxy’’:  true/false,
‘‘ai_logic_purge_accounts’’:  [ ‘‘List the exact account
numbers to be purged based on this sample.  If none, return
empty array.’’  ], ‘‘ai_logic_purge_keywords’’:  [ ‘‘List the
exact, lowercase words in the account_name column that you
identified and should be purged.  If none, return empty
array.’’  ], ‘‘rename_columns’’:  { ‘‘amount_including_vat’’:
‘‘amount_excluding_vat’’ } // Only include this key-value pair
if the Column Renaming Logic conditions are met.  Otherwise,
return null.  }
```

Appendix E — Quantitative Performance Analysis Table

File ID	Filetype	Classification	Size (MB)	Row Amount	Country	Manual Time	n8n Time	Human Review?
1	csv	Standard	23.1	48513	FI	07:58	01:18	
2	xlsx	Standard	5.47	90742	FI	02:48	03:58	
3	csv	Missing Must Column	9.87	23773	FI	06:03	06:03	Y, manual fallback
4	xlsx	Semantic Anomaly	1.83	26642	FI	02:00	00:49	
5	xlsx	Standard	0.8	9252	FI	01:45	00:40	
6	xlsx	Standard	2.9	43799	FI	02:07	01:35	
7	csv	Standard	3.3	20491	FI	05:23	00:42	
8	xlsx	Standard	3.62	80167	FI	01:18	02:59	
9	xlsx	Standard	9.65	91461	FI	02:22	05:00	
10	xlsx	Semantic Anomaly	1.47	23601	SE	03:03	00:39	
11	xlsx	Standard	1.01	15894	SE	01:10	00:31	
12	xlsx	Standard	2.64	50553	SE	02:20	01:34	
13	xlsx	Semantic Anomaly	2.29	40452	SE	02:55	01:20	
14	xlsx	Standard	6.28	136969	SE	01:27	08:54	
15	xlsx	Standard	0.14	1440	SE	01:03	00:08	
16	xlsx	Standard	13.8	25238	SE	01:35	01:20	
17	csv	Standard	1.8	15397	SE	00:49	00:31	
18	xlsx	Standard	0.02	297	SE	00:58	00:06	
19	xlsx	Semantic Anomaly	4.9	54356	SE	02:21	02:09	
20	xlsx	Semantic Anomaly	9.1	139505	SE	02:34	05:43	
					Manual Upload	01:50	/	
					Avg. Time	**02:41**	**02:18**	

**Note: For ID 3, execution halted, and SET converged to AAT due to HITL routing.*

**Note: n8n Total SET includes unattended machine processing times for large files.*

**Note: Batch Total manual time includes required manual upload processes.*